MATH 521

Analysis I

Midterm 2 Practice Workbook

University of Wisconsin–Madison

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Covers: Homework 5 – Homework 9 & Practice Midterm 2

Name: _____

Date: _____

Contents

How to Use This Workbook **2**

1 Homework 5 **3**
1.1 Problem 1 4
1.2 Problem 2 9
1.3 Problem 3 15
1.4 Problem 4 21

2 Homework 6 **27**
2.1 Problem 1 28
2.2 Problem 2 33
2.3 Problem 3 38
2.4 Problem 4 44
2.5 Problem 5 49
2.6 Problem 6 54

3 Homework 7 **60**
3.1 Problem 1 61
3.2 Problem 2 67
3.3 Problem 3 72
3.4 Problem 4 77
3.5 Problem 5 83
3.6 Problem 6 88

4 Homework 8 **92**
4.1 Problem 1 93
4.2 Problem 2 97
4.3 Problem 3 103
4.4 Problem 4 107
4.5 Problem 5 113
4.6 Problem 6 117

5 Homework 9 **121**
5.1 Problem 1 122
5.2 Problem 2 127
5.3 Problem 3 132
5.4 Problem 4 137
5.5 Problem 5 141
5.6 Problem 6 146

6 Practice Midterm 2 **150**

How to Use This Workbook

This workbook is designed for **active recall and deliberate practice**—not passive reading. Each problem is broken into **five passes** of increasing depth, followed by the official solution and a brief reflection.

Pass 1 **First Diagnosis.** Before writing anything, identify the topic, what is being asked, and which tools are likely relevant. Do this purely from memory.

Pass 2 **Proof Recipe Card.** Fill in the structured template: hypotheses, conclusion, key definitions, and a one-sentence strategy. Commit to a plan before writing.

Pass 3 **Self-Explanation.** Answer targeted questions about *why* the proof works. This is metacognition—one of the most effective study techniques known.

Pass 4 **Skeleton Proof.** Write the logical scaffold of the proof in the labeled blanks. Full sentences are not required yet; just the structure.

Pass 5 **Full Proof Attempt.** Write a complete, polished proof from scratch in the large blank space. *Only then* compare to the Official Solution below.

Suggested workflow: Complete Passes 1–3 first, then attempt Passes 4–5, then read the Official Solution. Use the Reflection section last to solidify understanding.

Homework 5

Problem 1

Problem Statement **Homework 5, Problem 1**

Let $\{a_n\}_{n\geq 1}$ be a sequence of positive real numbers. Prove that

$$\lim_{n\to\infty} a_n = +\infty \quad \text{if and only if} \quad \lim_{n\to\infty} \frac{1}{a_n} = 0.$$

Pass 1 – First Diagnosis

Topic: _____

What is being asked?: _____

Definitions likely relevant:

_____

Theorem / proof method likely relevant:

_____

What should the first line of the proof be?: _____

Pass 2 – Proof Recipe Card

Hypotheses: _____

Conclusion: _____

Definitions to unpack:

_____

Useful theorem / lemma / test: _____

Key inequality / identity: _____

One-sentence proof strategy:

_____

Pass 3 – Self-Explanation

Why does the chosen method apply here?: _____

What is the key move in the ($\Rightarrow$) direction?: _____

What is the key move in the ($\Leftarrow$) direction?: _____

What is the central inequality that drives the proof?: _____

Common mistake to avoid:: _____

Pass 4 – Skeleton Proof

($\Rightarrow$) Setup:

($\Rightarrow$) Main argument:

($\Rightarrow$) Conclusion:

($\Leftarrow$) Setup:

($\Leftarrow$) Main argument:

($\Leftarrow$) Conclusion:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 5 – Selected Solutions**

Problem 1. Let $\{a_n\}_{n\geq 1}$ be a sequence of positive real numbers. Prove that $\lim_{n\to\infty} a_n = +\infty$ if and only if $\lim_{n\to\infty} \frac{1}{a_n} = 0$.

Solution:

($\Rightarrow$) Fix $\epsilon > 0$. As $\lim a_n = +\infty$, then there exists N so that

$$a_n > \frac{1}{\epsilon} \quad \text{for all } n \geq N.$$

So, for $n \geq N$ we have

$$-\epsilon < 0 < \frac{1}{a_n} < \epsilon \implies \left|\frac{1}{a_n} - 0\right| < \epsilon.$$

Therefore $\lim \dfrac{1}{a_n} = 0$.

($\Leftarrow$) Fix $M > 0$. As $\lim \dfrac{1}{a_n} = 0$, then there exists N so that

$$\left|\frac{1}{a_n} - 0\right| < \frac{1}{M} \quad \text{for all } n \geq N.$$

Note that $\left|\dfrac{1}{a_n} - 0\right| = \dfrac{1}{a_n} > 0$. So

$$a_n > M \quad \text{for all } n \geq N.$$

Therefore $\lim a_n = +\infty$. $\square$

Reflection & Variation

What was the key definition used in each direction?: _____

Why was the hypothesis $a_n > 0$ essential? Where did it appear?: _____

What would change if $a_n > 0$ were replaced by $a_n \neq 0$? Where would the argument break?:

_____

What nearby exam variation could appear?: _____

Problem 2

Problem Statement **Homework 5, Problem 2**

Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that if $\{a_{k_n}\}_{n\geq 1}$ is a subsequence of $\{a_n\}_{n\geq 1}$ that converges to $L \in \mathbb{R}$, then

$$\liminf_{n\to\infty} a_n \;\leq\; L \;\leq\; \limsup_{n\to\infty} a_n.$$

(Hint: As $\{a_n\}_{n\geq 1}$ is not necessarily bounded, we have to consider the cases $\liminf_{n\to\infty} a_n = \pm\infty$ and $\limsup_{n\to\infty} a_n = \pm\infty$. However, the proof is much easier in these cases.)

Pass 1 – First Diagnosis

Topic: _____

What is being asked?: _____

Definitions likely relevant:

_____

Theorem / proof method likely relevant:

_____

Why does the proof split into cases?: _____

What should the first line of the proof be?: _____

Pass 2 – Proof Recipe Card

Hypotheses: _____

Conclusion (two inequalities): _____

Definitions to unpack ($\limsup$, $\liminf$, **subsequence**)**:**

_____

Useful theorem / lemma (hint: HW4): _____

What are the three cases for $\limsup a_n$?: _____

One-sentence proof strategy for $L \leq \limsup a_n$::

_____

Pass 3 – Self-Explanation

Why is the case $\limsup a_n = +\infty$ trivial?: _____

Why is the case $\limsup a_n = -\infty$ impossible?: _____

In the finite case, what key inequality holds for every $n \geq N$?: _____

What is the "engine" of the argument (which lemma closes it)?: _____

The solution says "using similar steps" for $\liminf$. What changes?: _____

Common mistake: why does the case analysis matter?: _____

Pass 4 – Skeleton Proof

We prove $L \leq \limsup_{n\to\infty} a_n$. *The inequality* $\liminf_{n\to\infty} a_n \leq L$ *follows by similar steps.*

Case 1: $\limsup a_n = +\infty$.

Case 2: $\limsup a_n = -\infty$ (show this is impossible).

Case 3: $\limsup a_n \in \mathbb{R}$.

Key inequality for $n \geq N$*:*

Apply HW4#3 to $\{a_{k_n}\}$*:*

Apply HW4#3 again to $v_N = \sup\{a_n : n \geq N\}$*:*

Conclusion:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 5 – Selected Solutions**

Problem 2. Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that if $\{a_{k_n}\}_{n\geq 1}$ is a subsequence of $\{a_n\}_{n\geq 1}$ that converges to $L \in \mathbb{R}$, then $\liminf\limits_{n\to\infty} a_n \leq L \leq \limsup\limits_{n\to\infty} a_n$.

Solution: Let's start with the second inequality: $L \leq \limsup_{n\to\infty} a_n$.

Case: $\limsup a_n = +\infty$. Then the statement is automatically true, since every real number L satisfies $L < +\infty$.

Case: $\limsup a_n = -\infty$. As $\liminf a_n \leq \limsup a_n$, then we must have $\liminf a_n = -\infty = \limsup a_n$. This means that $\lim a_n = -\infty$ by Lecture 14.
Fix $m < 0$. As $\lim a_n = -\infty$, there exists N so that

$$a_n < m \quad \text{for all } n \geq N.$$

As $k_n \geq n$ for all n, this implies

$$a_{k_n} < m \quad \text{for all } n \geq N.$$

Therefore $L = \lim a_{k_n} = -\infty$. This contradicts that $L \in \mathbb{R}$, and so this case is impossible.

Case: $\limsup a_n \in \mathbb{R}$. For $N \geq 1$, notice that

$$a_{k_n} \leq \sup\{a_n : n \geq N\} \quad \text{for all } n \geq N.$$

We know that the sequence $\{a_{k_n}\}$ converges to L. So, by HW4#3, we have

$$L = \lim a_{k_n} \leq \sup\{a_n : n \geq N\}.$$

On the other hand, $v_N = \sup\{a_n : n \geq N\}$ is a sequence that converges to $\limsup a_n$. So, using HW4#3 again, we conclude that

$$L \leq \lim_{N\to\infty} \sup\{a_n : n \geq N\} = \limsup a_n.$$

Exercise: Using similar steps, prove that $\liminf\limits_{n\to\infty} a_n \leq L$. □

Reflection & Variation

What result (from HW4) was the key lemma, and what does it say?: ____

Why was the case $\limsup a_n = -\infty$ impossible but not skipped?: _____

Carry out the "Exercise": sketch the proof that $\liminf a_n \leq L$, identifying exactly where the argument differs.:

What nearby exam variation could appear?: _____

Problem 3

Problem Statement **Homework 5, Problem 3**

Let $\{a_n\}_{n\geq 1}$ and $\{b_n\}_{n\geq 1}$ be two bounded sequences. Prove that

$$\limsup_{n\to\infty}(a_n + b_n) \;\leq\; \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

Pass 1 – First Diagnosis

Topic: _____

What is being asked?: _____

Definitions likely relevant:

_____

Theorem / proof method likely relevant:

_____

What should the first line of the proof be?: _____

Pass 2 – Proof Recipe Card

Hypotheses: _____

Conclusion: _____

Definitions to unpack ($\sup$ **of a set,** $\limsup$ **via tail sups):**

_____

Useful theorem / lemma (hint: HW4): _____

Key inequality relating $\sup\{a_n + b_n\}$ **to** $\sup\{a_n\}$ **and** $\sup\{b_n\}$**::** _____

One-sentence proof strategy:

_____

Pass 3 – Self-Explanation

Why does $a_n \leq \sup\{a_n : n \geq N\}$ hold for all $n \geq N$?: _____

Why does that imply $\sup\{a_n + b_n : n \geq N\} \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}$?: _____

Why does boundedness guarantee that all three $\limsup$s are finite?: _____

What is the role of HW4#3 at the very end of the proof?: _____

Why does the proof need boundedness? Where would it fail without it?: _____

Common mistake: confusing $\sup(A + B) \leq \sup A + \sup B$ with equality.: _____

Pass 4 – Skeleton Proof

Setup: Fix $N \geq 1$.

Pointwise inequality (for $n \geq N$):

Sup inequality (upper bound argument):

Convergence of each tail sup (use boundedness + lecture):

Apply HW4#3 to the difference sequence:

Conclusion:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 5 – Selected Solutions**

Problem 3. Let $\{a_n\}_{n\geq 1}$ and $\{b_n\}_{n\geq 1}$ be two bounded sequences. Prove that $\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n$.

Solution: Fix $N \geq 1$. For $n \geq N$ we have

$$a_n \leq \sup\{a_n : n \geq N\}, \qquad b_n \leq \sup\{b_n : n \geq N\},$$

and so

$$a_n + b_n \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

In other words, $\sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}$ is an upper bound for the set $\{a_n + b_n : n \geq N\}$. Therefore

$$\sup\{a_n + b_n : n \geq N\} \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

Altogether, we have shown that for any $N \geq 1$ we have

$$\sup\{a_n + b_n : n \geq N\} \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

As $\{a_n\}$ is a bounded sequence, we know from lecture that the increasing sequence $v_N = \sup\{a_n : n \geq N\}$ converges to $v = \limsup a_n \in \mathbb{R}$. Similarly, as $\{b_n\}$ is bounded, the sequence $\sup\{b_n : n \geq N\}$ converges to $\limsup b_n$ as $N \to \infty$. By the inequality above, we see that the sequence $\sup\{a_n+b_n : n \geq N\}$ is also bounded above. Therefore $\limsup(a_n + b_n)$ is a finite number, and $\sup\{a_n + b_n : n \geq N\}$ converges to $\limsup(a_n + b_n)$ as $N \to \infty$.
By the limit laws, the sequence

$$\sup\{a_n + b_n : n \geq N\} - \sup\{a_n : n \geq N\} - \sup\{b_n : n \geq N\}$$

converges as $N \to \infty$. We also know that it is ≤ 0 for all $N \geq 1$. Therefore, by HW4#3, the limit is also ≤ 0:

$$\limsup_{n\to\infty}(a_n + b_n) - \limsup_{n\to\infty} a_n - \limsup_{n\to\infty} b_n \leq 0$$
$$\implies \quad \limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

□

Reflection & Variation

What was the two-step structure of the proof (pointwise, then limit)?: ___
Why can equality fail in general? Give a quick example.: _____
Could you prove the analogous statement for $\liminf$**? What inequality di-**

rection would you get?:

_____

What nearby exam variation could appear?: _____

Problem 4

Problem Statement **Homework 5, Problem 4**

Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that

$$\limsup_{n\to\infty} |a_n| = 0 \quad \text{if and only if} \quad \lim_{n\to\infty} a_n = 0.$$

Pass 1 – First Diagnosis

Topic: _____

What is being asked?: _____

Definitions likely relevant:

_____

Theorem / proof method likely relevant:

_____

What should the first line of the proof be?: _____

Pass 2 – Proof Recipe Card

Hypotheses: _____

Conclusion: _____

Definitions to unpack ($\limsup$, $\liminf$, $\lim$)**:**

_____

Useful theorem / lemma (hint: Lecture 10, Lecture 14, HW4): _____

Key sandwich / squeeze in the ($\Rightarrow$) direction:: _____

One-sentence proof strategy for each direction:

_____

Pass 3 – Self-Explanation

Why is $\liminf |a_n| \geq 0$ **always true?:** _____

What does the sandwich $0 \leq \liminf |a_n| \leq \limsup |a_n| = 0$ **give you?:** _____

Which lecture result converts $\liminf = \limsup$ **into an ordinary limit?:** _____

Why does $||a_n| - 0| < \epsilon$ **imply** $|a_n - 0| < \epsilon$**?:** _____

In the ($\Leftarrow$) **direction, which earlier result lets you go from** $\lim a_n = 0$ **to** $\lim |a_n| = 0$**?:** _____

Common mistake: assuming $\limsup |a_n| = 0$ **directly gives** $|a_n| \to 0$ **without justification.:** _____

Pass 4 – Skeleton Proof

($\Rightarrow$) Suppose $\limsup |a_n| = 0$**.**

Claim: $\liminf |a_n| \geq 0$.

Sandwich:

Conclude $\liminf = \limsup$, *apply Lecture 10:*

Convert $\lim |a_n| = 0$ *to* $\lim a_n = 0$*:*

($\Leftarrow$) Suppose $\lim a_n = 0$**.**

Apply HW4#1:

Apply Lecture 14:

Conclusion:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 5 – Selected Solutions**

Problem 4. Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that $\limsup_{n\to\infty} |a_n| = 0$ if and only if $\lim_{n\to\infty} a_n = 0$.

Solution:

($\Rightarrow$) Suppose $\limsup |a_n| = 0$.
Claim: $\liminf |a_n| \geq 0$. As $|a_n| \geq 0$ for all n, we have

$$\inf_{n\geq N} |a_n| \geq 0 \quad \text{for all } N \geq 1.$$

Therefore

$$\lim_{N\to\infty} \inf_{n\geq N} |a_n| \geq 0$$

by HW4#3.
Altogether, we have

$$0 \leq \liminf |a_n| \leq \limsup |a_n| = 0.$$

Therefore

$$\liminf |a_n| = 0 = \limsup |a_n|.$$

By Lecture 10, this means that

$$\lim |a_n| = 0.$$

In other words: for all $\epsilon > 0$ there exists N so that

$$\big||a_n| - 0\big| < \epsilon \quad \text{for all } n \geq N.$$

This is the same as: for all $\epsilon > 0$ there exists N so that

$$|a_n - 0| < \epsilon \quad \text{for all } n \geq N.$$

So $\lim a_n = 0$.

($\Leftarrow$) Suppose $\lim a_n = 0$. By HW4#1, this implies

$$\lim |a_n| = 0.$$

By Lecture 14, we have

$$\liminf |a_n| = 0 = \limsup |a_n|.$$

So $\limsup |a_n| = 0$, as desired. $\square$

Reflection & Variation

What does Lecture 10 say, and why does it apply here?: _____

What does Lecture 14 say, and why does it apply in the ($\Leftarrow$) direction?: _

What would change if the statement were $\limsup a_n = 0 \iff \lim a_n = 0$? Is this even true? Construct a counterexample or prove it.:

_____

What nearby exam variation could appear?: _____

Homework 6

Problem 1

Problem Statement **Homework 6, Problem 1**

Prove that the series

$$\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$$

converges, and find its value. (Hint: You computed the partial sums on Homework 1.)

Pass 1 – First Diagnosis

Topic: _____

What two things are you asked to show?: _____

Definitions likely relevant:

_____

What does the hint suggest? What technique applies here?: _____

Partial fractions: how would you decompose $\frac{1}{(3n-2)(3n+1)}$?:

_____

What should the first line of the proof be?: _____

Pass 2 – Proof Recipe Card

Hypotheses: _____

Conclusion (convergence and value): _____

Partial fractions decomposition: $\frac{1}{(3n-2)(3n+1)} =$:

_____

Closed form for s_n after telescoping:: _____

Key limit needed to conclude:: _____

One-sentence proof strategy:

_____

Pass 3 – Self-Explanation

What is the partial fractions decomposition, and how do you verify it?: ___

Which terms cancel in the telescoping sum?: _____

What is the closed form for s_n?: _____

What limit do you evaluate to finish, and why does it exist?: _____

Common mistake: stating the value without proving s_n converges.: _____

Pass 4 – Skeleton Proof

Partial fractions decomposition:

nth partial sum s_n (telescope and simplify):

Evaluate $\lim_{n\to\infty} s_n$:

Conclusion (convergence + value):

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 6 – Selected Solutions**

Problem 1. Prove that $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$ converges, and find its value.

No official solution was provided in the Homework 6 Selected Solutions packet.

The hint states: "You computed the partial sums on Homework 1." This points to a partial fractions decomposition that produces a telescoping partial sum. Write and verify your own solution in Pass 5 above, then check it independently.

Reflection & Variation

What was the partial fractions decomposition?: _____

What did s_n simplify to after telescoping?: _____

What is the general strategy for proving convergence and finding the value of a telescoping series?:

_____

What nearby exam variation could appear?: _____

Problem 2

Problem Statement **Homework 6, Problem 2**

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be convergent series.

(a) Prove that for any $k \in \mathbb{R}$, the series $\sum_{n=1}^{\infty} ka_n$ converges and

$$\sum_{n=1}^{\infty} ka_n = k \sum_{n=1}^{\infty} a_n.$$

(b) Prove that the series $\sum_{n=1}^{\infty}(a_n + b_n)$ converges and

$$\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.$$

Pass 1 – First Diagnosis

Topic: _____

What is being asked in (a)? In (b)?: _____

Definitions likely relevant (partial sums, convergence of a series):

_____

Which limit law applies in (a)? In (b)?: _____

What should the first line of each proof be?: _____

Pass 2 – Proof Recipe Card

Hypotheses: _____

Conclusion for (a) / Conclusion for (b): _____

Let $s_n = \sum_{j=1}^{n} a_j$. The partial sums of $\sum ka_n$ equal:: _____

Let $t_n = \sum_{j=1}^{n} b_j$. The partial sums of $\sum(a_n + b_n)$ equal:: _____

Limit law used in (a):: _____

Limit law used in (b):: _____

One-sentence proof strategy (same idea for both parts):

_____

Pass 3 – Self-Explanation

Why do the partial sums of $\sum ka_n$ equal ks_n?: _____

Why do the partial sums of $\sum(a_n + b_n)$ equal $s_n + t_n$?: _____

What does it mean, by definition, for a series to converge to a number L?:

What limit law is being applied, and what does it say?: _____

Common mistake: confusing convergence of $\{a_n\}$ with convergence of $\sum a_n$.: _____

Pass 4 – Skeleton Proof

(a) Define s_n; identify partial sums of $\sum ka_n$:

(a) Apply limit law; conclude:

(b) Define s_n, t_n; identify partial sums of $\sum(a_n + b_n)$:

(b) Apply limit law; conclude:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 6 – Selected Solutions**

Problem 2. Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be convergent series.

Solution:

(a) Let $s_n = \sum_{j=1}^{n} a_j$ denote the nth partial sum. As $\sum a_n$ converges, we know that the sequence $\{s_n\}$ converges. Fix $k \in \mathbb{R}$. Then the partial sums $\sum_{j=1}^{n} ka_j$ are equal to ks_n. By the limit law for multiplication by a constant, we conclude that the sequence $\{ks_n\}_{n\geq 1}$ converges and $\lim(ks_n) = k \lim s_n$. By definition, this is exactly what it means for the series $\sum ka_n$ to converge to the number $k \sum a_n$.

(b) Let $s_n = \sum_{j=1}^{n} a_j$ and $t_n = \sum_{j=1}^{n} b_j$ denote the nth partial sums. As $\sum a_n$ and $\sum b_n$ converge, we know that the sequences $\{s_n\}$ and $\{t_n\}$ converge. Note that the partial sums $\sum_{j=1}^{n}(a_j + b_j)$ are equal to $s_n + t_n$. By the limit law for addition, we conclude that the sequence $\{s_n + t_n\}_{n\geq 1}$ converges and $\lim(s_n + t_n) = \lim s_n + \lim t_n$. By definition, this is exactly what it means for the series $\sum(a_n + b_n)$ to converge to the number $\sum a_n + \sum b_n$. $\square$

Reflection & Variation

What definition of series convergence was the key tool here?: _____

Does the result extend to $\sum(a_n - b_n)$? Why or why not?: _____

Can you use these results to show $\sum c_n$ diverges if it can be written as a linear combination of series one of which diverges?:

_____

What nearby exam variation could appear?: _____

Problem 3

Problem Statement **Homework 6, Problem 3**

(a) Prove that

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(Hint: $\frac{k-1}{2^{k+1}} = \frac{k}{2^k} - \frac{k+1}{2^{k+1}}$.)

(b) Use part (a) to compute $\sum_{n=1}^{\infty} \frac{n}{2^n}$.

Pass 1 – First Diagnosis

Topic: _____

What is being asked in (a)? In (b)?: _____

Definitions likely relevant (partial sums, induction, subsequences):

_____

What does the hint in (a) suggest about how to use the inductive step?: _____

How does part (b) reduce to part (a) via Problem 2?: _____

What should the first line of the proof of (a) be?: _____

Pass 2 – Proof Recipe Card

Hypotheses for (a): _____

Claim to prove by induction in (a): $s_n =$: _____

How is the hint used in the inductive step?: _____

Key limit needed ($\lim \frac{n}{2^n}$)**: prove it by:** _____

Algebraic identity used in (b): $\frac{k}{2^k} =$: _____

One-sentence proof strategy for (b):

_____

Pass 3 – Self-Explanation

What is the induction hypothesis, and what is the base case?: _____

How does the hint rewrite the new term $\frac{n}{2^{n+2}}$ in the inductive step?: _____

Why does $\frac{n+1}{2^{n+1}} \to 0$? (What earlier result is used?): _____

In (b), why does the algebraic identity allow you to apply Problem 2?: _____

Common mistake: not justifying $\lim \frac{n}{2^n} = 0$ separately.: _____

Pass 4 – Skeleton Proof

(a) Base case ($n = 1$):

(a) Inductive step (apply hint to rewrite new term):

(a) Prove $\lim \frac{n}{2^n} = 0$ (squeeze):

(a) Conclude $\lim s_n$ and state the series value:

(b) Write $\frac{k}{2^k}$ in terms of parts you know; apply Problem 2:

(b) Conclude:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Problem 3.

Solution:

(a) Let $s_n = \sum_{k=1}^{n} \frac{k-1}{2^{k+1}}$. We will prove that

$$s_n = \frac{1}{2} - \frac{n+1}{2^{n+1}} \quad \text{for } n \geq 1$$

by induction.

Base case: $s_1 = \dfrac{0}{4} = \dfrac{1}{2} - \dfrac{2}{4}$. ✓

Inductive step: Assume $s_n = \frac{1}{2} - \frac{n+1}{2^{n+1}}$ for some $n \geq 1$. Then

$$s_{n+1} = s_n + \frac{n}{2^{n+2}} = \frac{1}{2} - \frac{n+1}{2^{n+1}} + \frac{n+1}{2^{n+1}} - \frac{n+2}{2^{n+2}} = \frac{1}{2} - \frac{n+2}{2^{n+2}}.$$

(In the second equality we used the hint with $k = n+1$: $\frac{n}{2^{n+2}} = \frac{n+1}{2^{n+1}} - \frac{n+2}{2^{n+2}}$.) By induction, the formula holds for all $n \geq 1$.

Next, we claim that $\lim \dfrac{n}{2^n} = 0$. Notice that $n \leq \left(\dfrac{3}{2}\right)^n$ for all $n \geq 1$, and so

$$0 \leq \frac{n}{2^n} \leq \left(\frac{3/2}{2}\right)^n = \left(\frac{3}{4}\right)^n.$$

The right-hand side converges to zero as $n \to \infty$. Therefore $\frac{n}{2^n}$ also converges to zero by the squeeze theorem.

Altogether,

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \left(\frac{1}{2} - \frac{n+1}{2^{n+1}}\right) = \frac{1}{2}.$$

(We noted that $\frac{n+1}{2^{n+1}}$ is a subsequence of $\frac{n}{2^n}$, and therefore also converges to zero.) By definition of a convergent series, $\displaystyle\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}$.

(b) Note that

$$\sum_{k=1}^{n} \frac{k}{2^k} = 2\sum_{k=1}^{n} \frac{k-1}{2^{k+1}} + \sum_{k=1}^{n} \frac{1}{2^k}.$$

By part (a), $\sum \frac{k-1}{2^{k+1}}$ converges to $\frac{1}{2}$, so $2\sum \frac{k-1}{2^{k+1}}$ converges to 1 by Problem 2. From lecture, the geometric series $\sum_{k=1}^{\infty} \frac{1}{2^k}$ converges to $\frac{1}{1-(1/2)} - 1 = 1$. (We start at $k = 1$, not $k = 0$.) Together, $\displaystyle\sum_{k=1}^{\infty} \frac{k}{2^k}$ converges to $1 + 1 = 2$. □

Reflection & Variation

What was the key inductive formula, and where did the hint enter?: _____

Why was it enough to show $\lim \frac{n}{2^n} = 0$ **by squeeze with** $(3/4)^n$**?:** _____

In (b), identify which part of the solution used Problem 2 and which used the geometric series formula.:

_____

What nearby exam variation could appear?: _____

Problem 4

Problem Statement **Homework 6, Problem 4**

Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely and $\{b_n\}_{n\geq 1}$ is a bounded sequence, then the series $\sum_{n=1}^{\infty} a_n b_n$ also converges. (Hint: Use the Cauchy criterion.)

Pass 1 – First Diagnosis

Topic: _____

What is being asked?: _____

Definitions likely relevant (absolute convergence, bounded, Cauchy criterion for series):

_____

What does the hint suggest? State the Cauchy criterion for series.: _____

What should the first line of the proof be?: _____

Pass 2 – Proof Recipe Card

Hypotheses (two: absolute convergence of $\sum a_n$; $|b_n| \leq M$): _____

Conclusion: _____

Unpack the Cauchy criterion: what does it give for $\sum |a_n|$?:

_____

Key inequality bounding $\left|\sum_{k=m}^{n} a_k b_k\right|$:: _____

How do you choose ϵ' to make the bound $< \epsilon$?: _____

One-sentence proof strategy:

_____

Pass 3 – Self-Explanation

State the Cauchy criterion for series convergence from memory.: _____

Why does absolute convergence of $\sum a_n$ give Cauchy control on $\sum |a_k|$?: _____

How does $|b_n| \leq M$ enter the estimate?: _____

What is the central inequality chain: $\left|\sum_{k=m}^{n} a_k b_k\right| \leq \cdots$?: _____

Common mistake: forgetting that M is fixed, so dividing ϵ by M is valid.:

Pass 4 – Skeleton Proof

Setup: let M bound $\{b_n\}$; fix $\epsilon > 0$:

Apply Cauchy criterion to $\sum |a_n|$ (choose N):

Estimate $\left|\sum_{k=m}^{n} a_k b_k\right|$ for $n \geq m \geq N$:

Conclude by the Cauchy criterion:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 6 – Selected Solutions**

Problem 4. Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely and $\{b_n\}_{n\geq 1}$ is a bounded sequence, then $\sum_{n=1}^{\infty} a_n b_n$ also converges.

No official solution was provided in the Homework 6 Selected Solutions packet.

The hint states: "Use the Cauchy criterion." This points to estimating $\left|\sum_{k=m}^{n} a_k b_k\right|$ using the bound $|b_n| \leq M$ and the Cauchy condition on $\sum |a_k|$. Write and verify your own solution in Pass 5 above.

Reflection & Variation

State the Cauchy criterion for series. Why is it equivalent to convergence?:

Where exactly did boundedness of $\{b_n\}$ enter the argument?: _____

Does the result hold if $\sum a_n$ converges but not absolutely? Give a counterexample or prove it.:

_____

What nearby exam variation could appear?: _____

Problem 5

Problem Statement **Homework 6, Problem 5**

(a) Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then the series $\sum_{n=1}^{\infty} a_n^2$ also converges.
(b) Provide an example where $\sum_{n=1}^{\infty} a_n$ diverges and $\sum_{n=1}^{\infty} a_n^2$ converges.

Pass 1 – First Diagnosis

Topic: _____

What is being asked in (a)? In (b)?: _____

Definitions likely relevant (absolute convergence, comparison test):

_____

Key algebraic fact about x^2 vs. $|x|$ when $|x| < 1$:: _____

What should the first line of the proof of (a) be?: _____

For (b): what is the simplest candidate series?: _____

Pass 2 – Proof Recipe Card

Hypothesis for (a): _____

Conclusion for (a): _____

Key inequality (for $|x| \leq 1$): $x^2 \leq$: _____

Why does $a_n \to 0$ follow from absolute convergence?: _____

Why does $|a_n| < 1$ eventually, and what does that give you?: _____

Theorem used to conclude $\sum a_n^2$ converges:: _____

Example for (b): $a_n =$ Verify: $\sum a_n$ diverges because $\sum a_n^2$ converges because:

_____

Pass 3 – Self-Explanation

Why does $x^2 \leq |x|$ hold for $x \in [-1, 1]$? (Draw the graphs mentally.): _____

How does absolute convergence give $a_n \to 0$, and then $|a_n| < 1$ eventually?:

What is the comparison: $a_n^2 \leq$ _____ for all $n \geq N$?: _____
Why does adding back finitely many terms not affect convergence?: _____
For (b): why does the example work? Quote the lecture results used.: ___

Pass 4 – Skeleton Proof

(a) State the key inequality ($x^2 \leq |x|$ on $[-1, 1]$):

(a) Use $a_n \to 0$ to find N with $|a_n| < 1$:

(a) Apply comparison test to $\sum_{n=N}^{\infty} a_n^2$:

(a) Extend to full series; conclude:

(b) State example and verify both claims:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 6 – Selected Solutions**

Problem 5.

Solution:

(a) Note that $x^2 \leq |x|$ for any $x \in [-1, 1]$. (I recommend that you draw the graphs of x^2 and $|x|$ to check that this is true.)
As $\sum a_n$ converges absolutely, we have $\lim a_n = 0$, and so there exists an N so that $|a_n| < 1$ for all $n \geq N$. By the previous paragraph, this means that $a_n^2 \leq |a_n|$ for all $n \geq N$. As $\sum |a_n|$ converges, then $\sum_{n=N}^{\infty} a_n^2$ also converges by the comparison test. Adding the first $N-1$ terms back in, we conclude $\sum_{n=1}^{\infty} a_n^2$ converges as well.

(b) Consider $a_n = \dfrac{1}{n}$. In lecture, we proved that $\sum \dfrac{1}{n}$ diverges and $\sum \dfrac{1}{n^2}$ converges.$\square$

Reflection & Variation

Why does the proof need $|a_n| < 1$? Where exactly does it enter?: _____

Is the converse of (a) true: if $\sum a_n^2$ converges, must $\sum a_n$ converge absolutely?: _____

Could you weaken the hypothesis to "$\sum a_n$ converges (not necessarily absolutely)"? Where does the proof break?:

_____

What nearby exam variation could appear?: _____

Problem 6

Problem Statement **Homework 6, Problem 6**

For each of the following series, determine if it converges and prove your answer.

$$\text{(a)} \quad \sum_{n=2}^{\infty} \frac{1}{[n+(-1)^n]^2} \qquad \text{(b)} \quad \sum_{n=1}^{\infty} \left(\sqrt{n+1} - \sqrt{n}\right) \qquad \text{(c)} \quad \sum_{n=1}^{\infty} (-1)^n$$

Pass 1 – First Diagnosis

Topic: _____

Verdict for (a) (converges / diverges):: _____

Verdict for (b) (converges / diverges):: _____

Verdict for (c) (converges / diverges):: _____

Test / method for (a) / (b) / (c)::

_____

What should the first line of each proof be?: _____

Pass 2 – Proof Recipe Card

(a) Bound: $\frac{1}{[n+(-1)^n]^2} \leq$ _____ **(why does** $n + (-1)^n \geq n - 1$**?):** _____

(a) Comparison series (known to converge):: _____

(b) Closed form for $s_n = \sum_{k=1}^{n} (\sqrt{k+1} - \sqrt{k})$**::** _____

(b) Why does $s_n \to +\infty$**?:** _____

(c) Key theorem: if $\sum a_n$ **converges, then** $\lim a_n =$**:** _____

(c) Why does $(-1)^n \not\to 0$**?:** _____

Pass 3 – Self-Explanation

(a) Why is $n + (-1)^n \geq n - 1 \geq 1$ **for** $n \geq 2$**?:** _____

(a) What is the comparison series, and what makes it convergent?: _____

(b) How do you prove the closed form $s_n = \sqrt{n+1} - 1$ **by induction?:** _____

(b) What is the proof that $s_n \to +\infty$ **(give the** N **in terms of** M**)?:** _____

(c) What is Lecture 16's result, and how does it give the contradiction?: _

Common mistake in (c): thinking divergence of terms implies divergence of series (wrong direction).: _____

Pass 4 – Skeleton Proof

(a) State bound and comparison series; apply comparison test:

(b) Claim $s_n = \sqrt{n+1} - 1$; induction base case:

(b) Inductive step:

(b) Show $s_n \to +\infty$ (given $M > 0$, choose N):

(c) Contradiction argument (Lecture 16 + Lecture 10):

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 6 – Selected Solutions**

Problem 6. For each of the following series, determine if it converges and prove your answer.

Solution:

(a) We will prove that this series **converges**. For $n \geq 2$, we have

$$\frac{1}{[n+(-1)^n]^2} \leq \frac{1}{(n-1)^2}.$$

From lecture, we know that the series $\displaystyle\sum_{n=2}^{\infty} \frac{1}{(n-1)^2} = \sum_{k=1}^{\infty} \frac{1}{k^2}$ converges. Therefore our series $\sum \frac{1}{[n+(-1)^n]^2}$ also converges by the comparison test.

(b) We will prove that this series **diverges**. Let $s_n = \sum_{k=1}^{n}(\sqrt{k+1} - \sqrt{k})$ denote the nth partial sum.
Claim: $s_n = \sqrt{n+1} - 1$ for all $n \geq 1$. We argue by induction.

- *Base case:* For $n = 1$, $s_1 = \sqrt{2} - \sqrt{1} = \sqrt{2} - 1$. ✓
- *Inductive step:* Suppose $s_n = \sqrt{n+1} - 1$ for some $n \geq 1$. Then

$$\begin{aligned} s_{n+1} &= s_n + (\sqrt{n+2} - \sqrt{n+1}) \\ &= (\sqrt{n+1} - 1) + (\sqrt{n+2} - \sqrt{n+1}) = \sqrt{n+2} - 1. \end{aligned}$$

By induction, $s_n = \sqrt{n+1} - 1$ for all $n \geq 1$.
Notice that $s_n = \sqrt{n+1} - 1$ diverges to $+\infty$. Indeed, given $M > 0$, pick $N \in \mathbb{N}$ so that $N > (M+1)^2$. Then for any $n \geq N$:

$$\sqrt{n+1} - 1 \geq \sqrt{N} - 1 > \sqrt{(M+1)^2} - 1 = M.$$

As $\{s_n\}$ diverges, the series $\sum(\sqrt{n+1} - \sqrt{n})$ diverges.

(c) We will prove that this series **diverges**. Suppose towards a contradiction that $\sum(-1)^n$ converges. Then $\lim(-1)^n = 0$ by Lecture 16. However, we know that the sequence $\{(-1)^n\}$ diverges by Lecture 10, and so this is a contradiction. □

Reflection & Variation

(a) What is the comparison test, and why is the bound $\frac{1}{(n-1)^2}$ valid?: _____

(b) The partial sums telescope. Why does a telescoping series diverge here?: _____

(c) State Lecture 16's result ("necessary condition for convergence") precisely.: _____

What is the contrast between (a) and (b)? Both involve terms that shrink. Why does one converge and the other diverge?:

_____

What nearby exam variation could appear?: _____

Homework 7

Problem 1

Problem Statement **Homework 7, Problem 1**

For each of the following series, determine if it converges and prove your answer.

$$\text{(a)} \sum_{n=2}^{\infty} \frac{1}{\log n} \qquad \text{(b)} \sum_{n=1}^{\infty} \frac{n-1}{n^2} \qquad \text{(c)} \sum_{n=1}^{\infty} \frac{n^2}{n!}$$

Pass 1 – First Diagnosis

Topic: _____

Verdict for (a):: _____

Verdict for (b):: _____

Verdict for (c):: _____

Test / method for each part::

_____

What should the first line of each proof be?: _____

Pass 2 – Proof Recipe Card

(a) Test used: Key comparison / bound:: _____

(a) Series you compare to, and why it diverges:: _____

(b) Key inequality: for $n \geq 2$, $\frac{n-1}{n^2} \geq$**:** _____

(b) Series you compare to, and why it diverges:: _____

(c) Compute $\left|\frac{a_{n+1}}{a_n}\right|$ **for** $a_n = \frac{n^2}{n!}$**::** _____

(c) $\lim \left|\frac{a_{n+1}}{a_n}\right| =$ **Conclusion::** _____

Pass 3 – Self-Explanation

(a) What is the dyadic criterion, and how does it apply here?: _____

(a) Why does $2^n a_{2^n} \geq \frac{1}{\log 2} \cdot \frac{2^n}{n+1}$ **and why does this diverge?:** _____

(b) Why does $n - 1 \geq n/2$ **for** $n \geq 2$**? What does this give for** $\frac{n-1}{n^2}$**?:** _____

(c) Why does $\left|\frac{a_{n+1}}{a_n}\right| \to 0$? Which step in the simplification is key?: _____

Common mistake: applying ratio test to (a) — the limit is 1, so the test is inconclusive.: _____

Pass 4 – Skeleton Proof

(a) State a_k for dyadic criterion; verify $\{a_k\}$ is decreasing and nonneg:

(a) Compute $2^n a_{2^n}$; show it diverges to $+\infty$:

(a) Conclude:

(b) State key inequality for $n \geq 2$:

(b) Name comparison series; conclude:

(c) Compute $\left|\frac{a_{n+1}}{a_n}\right|$ and simplify:

(c) State $\limsup \left|\frac{a_{n+1}}{a_n}\right|$; conclude by ratio test:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 7 – Solutions**

Problem 1.

Solution:

(a) We'll use the dyadic criterion. Consider the series $\sum_{k=1}^{\infty} a_k$ with $a_k = \frac{1}{\log(k+1)}$ for $k \geq 1$. (Strictly speaking, in order to apply the dyadic criterion we need a series that starts at $n = 1$, so we reindexed.)
Then $\{a_k\}_{k\geq 1}$ is a decreasing sequence of nonnegative numbers. Therefore, by the dyadic criterion, $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{n=0}^{\infty} 2^n a_{2^n}$ converges. Note that

$$2^n a_{2^n} = \frac{2^n}{\log(2^n+1)} \geq \frac{2^n}{\log(2^{n+1})} = \frac{1}{\log 2} \cdot \frac{2^n}{n+1}.$$

Now, the sequence $\frac{2^n}{n+1}$ diverges to $+\infty$, and so it cannot converge to 0. Multiplying by the constant $\frac{1}{\log 2}$, we see that $2^n a_{2^n}$ also cannot converge to 0. Therefore the series $\sum 2^n a_{2^n}$ diverges, and thus $\sum a_k$ **diverges**.

(b) We'll use the comparison test. For $n \geq 2$, we have

$$n - 1 \geq n - \frac{n}{2} = \frac{n}{2} \implies \frac{n-1}{n^2} \geq \frac{n/2}{n^2} = \frac{1}{2n}.$$

From lecture we know that $\sum \frac{1}{n}$ diverges, and thus $\sum_{n=2}^{\infty} \frac{1}{2n}$ diverges as well. Therefore, by the comparison test, $\sum_{n=2}^{\infty} \frac{n-1}{n^2}$ also **diverges**. (Note that $\frac{n-1}{n^2} = 0$ for $n = 1$, so $\sum_{n=2}^{\infty}$ and $\sum_{n=1}^{\infty}$ are the same series.)

(c) We'll use the ratio test. Set $a_n = \frac{n^2}{n!}$. Then

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)^2 \cdot n!}{n^2 \cdot (n+1)!} = \frac{(n+1)^2}{n^2(n+1)} = \frac{n+1}{n^2} \to 0 \quad \text{as } n \to \infty.$$

Therefore

$$\limsup \left|\frac{a_{n+1}}{a_n}\right| = \lim \left|\frac{a_{n+1}}{a_n}\right| = 0 < 1,$$

and so the series **converges** by the ratio test. □

Reflection & Variation

State the dyadic criterion precisely. Why does it apply in (a)?: _____

Why does the ratio test fail on (a)? What is $\lim |a_{n+1}/a_n|$ and why is that inconclusive?: _____

In (b), why was it sufficient to find a lower bound by $\frac{1}{2n}$ rather than trying the ratio test?:

What nearby exam variation could appear?: _____

Problem 2

Problem Statement **Homework 7, Problem 2**

For each of the following series, determine if it converges and prove your answer.

$$\text{(a)} \sum_{n=1}^{\infty} \frac{n^4}{4^n} \qquad \text{(b)} \sum_{n=1}^{\infty} \frac{n!}{n^4+3} \qquad \text{(c)} \sum_{n=1}^{\infty} \frac{2^n}{n!}$$

Pass 1 – First Diagnosis

Topic: _____

Verdict for (a):: _____

Verdict for (b):: _____

Verdict for (c):: _____

Test / method for each part::

_____

What should the first line of each proof be?: _____

Pass 2 – Proof Recipe Card

(a) Compute $\left|\frac{a_{n+1}}{a_n}\right|$ **for** $a_n = \frac{n^4}{4^n}$**::** _____

(a) $\lim \left|\frac{a_{n+1}}{a_n}\right| =$ **Conclusion::** _____

(b) Strategy: show terms $\frac{n!}{n^4+3}$ **do not converge to 0. Why?:** _____

(b) Key bound: for $n \geq 5$, $\frac{n^4+3}{n!} \leq$ _____, **which** $\to 0$, **so** $\frac{n!}{n^4+3} \to$**:** _____

(c) Compute $\left|\frac{a_{n+1}}{a_n}\right|$ **for** $a_n = \frac{2^n}{n!}$**::** _____

(c) $\lim \left|\frac{a_{n+1}}{a_n}\right| =$ **Conclusion::** _____

Pass 3 – Self-Explanation

(a) Where does the $\frac{(n+1)^4}{4n^4} \to \frac{1}{4}$ **come from?:** _____

(b) Why does $n! \geq n(n-1)(n-2)(n-3)(n-4)$ **for** $n \geq 5$**?:** _____

(b) How does HW5#1 convert $\frac{n^4+3}{n!} \to 0$ into $\frac{n!}{n^4+3} \to +\infty$?: _____

(c) Why does $\frac{2}{n+1} \to 0$? What does that say about $\limsup \left|\frac{a_{n+1}}{a_n}\right|$?: _____

Common mistake in (b): trying to apply ratio or root test instead of recognizing the necessary condition for convergence fails.: _____

Pass 4 – Skeleton Proof

(a) Ratio test — compute and simplify $|a_{n+1}/a_n|$:

(a) State $\limsup$; conclude:

(b) Show $\frac{n^4+3}{n!} \to 0$ for $n \geq 5$ (bound denominator from below):

(b) Apply HW5#1 to conclude $\frac{n!}{n^4+3} \to +\infty$; use necessary condition:

(c) Ratio test — compute and simplify $|a_{n+1}/a_n|$:

(c) State $\limsup$; conclude:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Problem 2.

Solution:

(a) We'll use the ratio test. Set $a_n = \frac{n^4}{4^n}$. Then

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)^4 \cdot 4^n}{n^4 \cdot 4^{n+1}} = \frac{(n+1)^4}{4n^4} \to \frac{1}{4} \quad \text{as } n \to \infty.$$

Therefore $\limsup \left|\frac{a_{n+1}}{a_n}\right| = \frac{1}{4} < 1$, so the series **converges** by the ratio test.

(b) This series **diverges**. In brief, the sequence $\frac{n!}{n^4+3}$ diverges to $+\infty$, and so it cannot converge to zero. More concretely, notice that for $n \geq 5$ we have

$$n! \geq n(n-1)(n-2)(n-3)(n-4),$$

and so

$$\frac{n^4+3}{n!} \leq \frac{n^4+3}{n(n-1)(n-2)(n-3)(n-4)} \to 0 \quad \text{as } n \to \infty.$$

Therefore the sequence $\frac{n!}{n^4+3} \to +\infty$ by HW5#1. In particular, there must exist an N so that $\frac{n!}{n^4+3} \geq 1$ for all $n \geq N$, and so $\frac{n!}{n^4+3}$ cannot converge to 0.

(c) We'll use the ratio test. Set $a_n = \frac{2^n}{n!}$. Then

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{2^{n+1} \cdot n!}{(n+1)! \cdot 2^n} = \frac{2}{n+1} \to 0 \quad \text{as } n \to \infty.$$

Therefore $\limsup \left|\frac{a_{n+1}}{a_n}\right| = 0 < 1$, so the series **converges** by the ratio test. □

Reflection & Variation

State the ratio test precisely: what are the three cases?: _____

In (b), what is the "necessary condition for convergence," and why does it apply?: _____

Compare (a) and (c): both converge via ratio test. What structural feature of these series makes ratio test work well?:

_____

What nearby exam variation could appear?: _____

Problem 3

Problem Statement **Homework 7, Problem 3**

For each of the following series, determine if it converges and prove your answer.

$$\text{(a)} \sum_{n=1}^{\infty} \frac{1}{n^n} \qquad \text{(b)} \sum_{n=1}^{\infty} \frac{n!}{n^n} \qquad \text{(c)} \sum_{n=1}^{\infty} \frac{(-1)^n n!}{2^n}$$

Pass 1 – First Diagnosis

Topic: _____

Verdict for (a):: _____

Verdict for (b):: _____

Verdict for (c):: _____

Test / method for each part (note: all three use the root or ratio test)::

_____

What makes (b) require a Lecture 15 result?: _____

Pass 2 – Proof Recipe Card

(a) Compute $|a_n|^{1/n}$ **for** $a_n = \frac{1}{n^n}$**::** _____

(a) $\limsup |a_n|^{1/n} =$ **Conclusion::** _____

(b) Compute $|a_n|^{1/n}$ **for** $a_n = \frac{n!}{n^n}$**::** _____

(b) Lecture 15 says $\frac{(n!)^{1/n}}{n} \to$ **Conclusion::** _____

(c) Strategy: show $\frac{(-1)^n n!}{2^n}$ **does not converge to 0.:** _____

(c) Key: $\frac{2^n}{n!} \to 0$ **(from Problem 2c), so by HW5#1,** $\frac{n!}{2^n} \to$**:** _____

Pass 3 – Self-Explanation

(a) Why does $|a_n|^{1/n} = 1/n \to 0$ **immediately give** $\limsup = 0 < 1$**?:** _____

(b) What did Lecture 15 show about $(n!)^{1/n}/n$**, and how does it enter the root test?:** _____

(c) How does convergence of $\sum \frac{2^n}{n!}$ **(Problem 2c) imply** $\frac{2^n}{n!} \to 0$**?:** _____

(c) Then how does HW5#1 give $\frac{n!}{2^n} \to +\infty$?: _____

Common mistake: trying ratio test on (b) and getting confused — root test is cleaner.: _____

Pass 4 – Skeleton Proof

(a) Root test — compute $|a_n|^{1/n}$; state $\limsup$; conclude:

(b) Root test — compute $|a_n|^{1/n} = (n!/n^n)^{1/n}$:

(b) Apply Lecture 15 to get the limit; state $\limsup$; conclude:

(c) Use Problem 2c to get $\frac{2^n}{n!} \to 0$; apply HW5#1:

(c) Conclude terms $\frac{(-1)^n n!}{2^n}$ do not converge to 0; series diverges:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Problem 3.

Solution:

(a) We'll use the root test. Set $a_n = \frac{1}{n^n}$. Then

$$|a_n|^{1/n} = \frac{1}{n} \to 0 \quad \text{as } n \to \infty.$$

Therefore $\limsup |a_n|^{1/n} = 0 < 1$, so the series **converges** by the root test.

(b) We'll use the root test. Set $a_n = \frac{n!}{n^n}$, and consider the sequence

$$|a_n|^{1/n} = \left(\frac{n!}{n^n}\right)^{1/n} = \frac{(n!)^{1/n}}{n}.$$

In Lecture 15 we showed that this sequence converges to $1/e$. Therefore $\limsup |a_n|^{1/n} = \frac{1}{e} < 1$, so the series **converges** by the root test.

(c) This series **diverges**, because the sequence $\frac{(-1)^n n!}{2^n}$ does not converge to 0. Specifically, we know that the sequence $\frac{2^n}{n!}$ converges to 0. (Indeed, we proved in Problem 2c that the series $\sum \frac{2^n}{n!}$ converges, and so its terms must converge to 0.) So, by HW5#1, the sequence $\frac{n!}{2^n}$ diverges to $+\infty$. In particular, there exists an N so that $\frac{n!}{2^n} \geq 1$ for all $n \geq N$. Therefore $\left|\frac{(-1)^n n!}{2^n}\right| = \frac{n!}{2^n} \geq 1$ for all $n \geq N$, so the terms $\frac{(-1)^n n!}{2^n}$ in our series cannot converge to 0. $\square$

Reflection & Variation

State the root test precisely: what are the three cases?: _____

Why is the root test more natural than the ratio test for (a) and (b)?: ___

In (c), the argument used HW5#1 (a result about $\lim a_n = +\infty$ iff $\lim 1/a_n = 0$). Identify exactly where it was applied.:

_____

What nearby exam variation could appear?: _____

Problem 4

Problem Statement **Homework 7, Problem 4**

Consider the series

$$\sum_{n=1}^{\infty} 2^{(-1)^n - n} = \frac{1}{4} + \frac{1}{2} + \frac{1}{16} + \frac{1}{8} + \frac{1}{64} + \frac{1}{32} + \cdots$$

(a) Show that the ratio test is inconclusive.
(b) Prove that the series converges using the root test.

Pass 1 – First Diagnosis

Topic: _____

What is $a_n = 2^{(-1)^n - n}$ explicitly for n even? For n odd?: _____

What does "ratio test is inconclusive" mean precisely?: _____

Why does the alternating structure cause the ratio test to fail here?: _____

What Lecture 15 fact will be needed for the root test?: _____

Pass 2 – Proof Recipe Card

(a) Write out first few values of $\left|\frac{a_{n+1}}{a_n}\right|$:: _____

(a) $\liminf \left|\frac{a_{n+1}}{a_n}\right| =$ $\limsup \left|\frac{a_{n+1}}{a_n}\right| =$**:** _____

(a) Why does $\liminf \leq 1 \leq \limsup$ **make the test inconclusive?:** _____

(b) For n even: $|a_n|^{1/n} = (2^{n-1})^{-1/n} =$ _____ $\to$**:** _____

(b) For n odd: $|a_n|^{1/n} = (2^{n+1})^{-1/n} =$ _____ $\to$**:** _____

(b) $\lim |a_n|^{1/n} =$ **Conclusion::** _____

Pass 3 – Self-Explanation

(a) The even and odd terms of $|a_{n+1}/a_n|$ are $\frac{1}{8}$ and 2 respectively. Verify this.: _____

(a) Recall: the ratio test is inconclusive when .....: _____

(b) For n even: write $a_n = 2^{1-n}$ and simplify $|a_n|^{1/n}$ step by step.: _____

(b) For n odd: write $a_n = 2^{-1-n}$ and simplify $|a_n|^{1/n}$ step by step.: _____

(b) What Lecture 15 fact is used to conclude $2^{1/n} \to 1$ and $2^{-1/n} \to 1$?: _____

Key lesson: when the ratio test fails due to oscillation, the root test may still work.: _____

Pass 4 – Skeleton Proof

(a) Write out $|a_{n+1}/a_n|$ for the first few n:

(a) Identify $\liminf$ and $\limsup$; state why test is inconclusive:

(b) For n even — compute $|a_n|^{1/n}$ and take $n \to \infty$:

(b) For n odd — compute $|a_n|^{1/n}$ and take $n \to \infty$:

(b) Combine: state $\lim |a_n|^{1/n}$; conclude by root test:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Problem 4.

Solution:

(a) Set $a_n = 2^{(-1)^n - n}$. Then the sequence $\left|\frac{a_{n+1}}{a_n}\right|$ is

$$\frac{1/2}{1/4} = 2, \quad \frac{1/16}{1/2} = \frac{1}{8}, \quad \frac{1/8}{1/16} = 2, \quad \frac{1/64}{1/8} = \frac{1}{8}, \quad \ldots$$

In other words, the even terms of this sequence are $\frac{1}{8}$ and the odd terms are 2. Therefore

$$\liminf \left|\frac{a_{n+1}}{a_n}\right| = \frac{1}{8}, \qquad \limsup \left|\frac{a_{n+1}}{a_n}\right| = 2,$$

and so

$$\liminf \left|\frac{a_{n+1}}{a_n}\right| \leq 1 \leq \limsup \left|\frac{a_{n+1}}{a_n}\right|.$$

Hence the ratio test is inconclusive.

(b) Set $a_n = 2^{(-1)^n - n}$. For n even, $a_n = 2^{1-n}$, so

$$|a_n|^{1/n} = \frac{1}{(2^{n-1})^{1/n}} = \frac{1}{2^{1-1/n}} = \frac{1}{2} \cdot 2^{1/n} \to \frac{1}{2} \quad \text{as } n \to \infty.$$

For n odd, $a_n = 2^{-1-n}$, so

$$|a_n|^{1/n} = \frac{1}{(2^{n+1})^{1/n}} = \frac{1}{2^{1+1/n}} = \frac{1}{2} \cdot \frac{1}{2^{1/n}} \to \frac{1}{2} \quad \text{as } n \to \infty.$$

(Recall from Lecture 15 that $r^{1/n} \to 1$ for any $r > 0$.) Therefore

$$\lim_{n\to\infty} |a_n|^{1/n} = \frac{1}{2} < 1,$$

and so the series converges by the root test. □

Reflection & Variation

Precisely: when is the ratio test "inconclusive"? Quote the exact condition.: _____

Why does oscillation in $|a_{n+1}/a_n|$ cause the ratio test to fail here?: _____

The root test gave $\limsup |a_n|^{1/n} = 1/2$ despite the oscillation. Why is the root test more robust to oscillation than the ratio test?:

What nearby exam variation could appear?: _____

Problem 5

Problem Statement **Homework 7, Problem 5**

Below are two series $\sum_{n=1}^{\infty} a_n$ and the numbers $s \in \mathbb{R}$ to which they converge. For each of these series, what is the set of all numbers $\tilde{s} \in \mathbb{R}$ such that there exists a rearrangement that satisfies $\sum_{n=1}^{\infty} \tilde{a}_n = \tilde{s}$? Prove your answer.

$$\text{(a)} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} = -\log 2 \qquad \text{(b)} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}$$

(Hint: Your proof should be quite short, because we already did most of the work in lecture.)

Pass 1 – First Diagnosis

Topic: _____

Does $\sum(-1)^n/n$ converge absolutely? Why or why not?: _____

Does $\sum(-1)^n/n^2$ converge absolutely? Why or why not?: _____

What two lecture theorems are relevant (one for conditionally convergent, one for absolutely convergent)?: _____

What is the answer for (a)? For (b)?: _____

Pass 2 – Proof Recipe Card

(a) Check: $\sum|(-1)^n/n|$ converges or diverges?: _____

(a) Check: $\sum(-1)^n/n$ converges? By what test?: _____

(a) Theorem applied (Riemann rearrangement): if $\sum a_n$ converges cond'ly, then...: _____

(a) Answer: set of achievable rearrangement sums =: _____

(b) Check: $\sum|(-1)^n/n^2|$ converges or diverges?: _____

(b) Theorem applied (absolute convergence): if $\sum a_n$ converges absolutely, then any rearrangement...: _____

(b) Answer: set of achievable rearrangement sums =: _____

Pass 3 – Self-Explanation

State the alternating series test precisely. Why does it apply in (a) and (b)?: _____

(a) The Riemann rearrangement theorem requires two conditions. What are they?: _____

(b) Why does absolute convergence force every rearrangement to the same sum?: _____

What is the contrast between (a) and (b)? What changes the answer from $\mathbb{R}$ to a singleton?: _____

Common mistake: confusing convergence with absolute convergence.: ____

Pass 4 – Skeleton Proof

(a) Verify $\sum(-1)^n/n$ converges (state test):

(a) Verify $\sum(-1)^n/n$ does NOT converge absolutely:

(a) Apply Riemann rearrangement theorem; state answer:

(b) Verify $\sum(-1)^n/n^2$ converges absolutely:

(b) Apply absolute convergence rearrangement theorem; state answer:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 7 – Solutions**

Problem 5.

Solution:

(a) We claim that the set is $\mathbb{R}$. Note that:

- $\sum \frac{(-1)^n}{n}$ converges: by the alternating series test, since $\left\{\frac{1}{n}\right\}$ is a decreasing sequence that converges to 0.
- $\sum \frac{(-1)^n}{n}$ does not converge absolutely: since $\left|\frac{(-1)^n}{n}\right| = \frac{1}{n}$, and we know $\sum \frac{1}{n}$ diverges by Lecture 11.

Fix $\tilde{s} \in \mathbb{R}$. Applying the Riemann rearrangement result (with $\alpha = \tilde{s} = \beta$), we deduce that there exists a rearrangement $\sum \tilde{a}_n$ that converges to $\tilde{s}$. This works for any $\tilde{s} \in \mathbb{R}$, so $\mathbb{R}$ is the desired set.

(b) We claim that the set is $\left\{-\frac{\pi^2}{12}\right\}$. Note that $\sum \frac{(-1)^n}{n^2}$ converges absolutely, since $\left|\frac{(-1)^n}{n^2}\right| = \frac{1}{n^2}$ and we know $\sum \frac{1}{n^2}$ converges by Lecture 16. Suppose that $\sum \tilde{a}_n$ is a rearrangement that converges to some $\tilde{s} \in \mathbb{R}$. Applying the second theorem from Lecture 20, we deduce that $\tilde{s} = -\frac{\pi^2}{12}$. This works for any rearrangement, so $-\frac{\pi^2}{12}$ is the only value that can be attained. $\square$

Reflection & Variation

State the Riemann rearrangement theorem precisely (two conditions, two conclusions).: _____

State the absolute convergence rearrangement theorem precisely.: _____

The series in (a) converges to $-\log 2$. Is $-\log 2$ itself in the set $\mathbb{R}$? (Of course — but what rearrangement gives it?):

_____

What nearby exam variation could appear?: _____

Problem 6

Problem Statement **Homework 7, Problem 6**

Prove or disprove the following statements:
(a) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_{2n}$ converges.
(b) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_{2n}$ converges absolutely.

Pass 1 – First Diagnosis

Topic: _____

Verdict for (a): true or false?: _____

Verdict for (b): true or false?: _____

For (a): what counterexample series comes to mind?: _____

For (b): what proof method does the hint in HW6#4 suggest?: _____

Pass 2 – Proof Recipe Card

(a) Counterexample: $a_n =$ **Verify** $\sum a_n$ **converges because::** _____

(a) Verify $\sum a_{2n}$ **diverges because::** _____

(b) Hypothesis: $\sum |a_n|$ **converges. Cauchy criterion gives:** $\exists N$ **s.t.:** _____

(b) Key bound: for $n \geq m \geq N$, $\sum_{k=m}^{n} |a_{2k}| \leq \sum_{j=2m}^{2n} |a_j| <$**:** _____

(b) Conclude by Cauchy criterion:: _____

Pass 3 – Self-Explanation

(a) Why does $\sum a_{2n} = \sum \frac{1}{2n}$ **diverge?:** _____

(b) State the Cauchy criterion for series. Why does it apply here?: _____

(b) Why is $\sum_{k=m}^{n} |a_{2k}| \leq \sum_{j=2m}^{2n} |a_j|$**? What terms are being added?:** _____

(b) Why does $\sum_{j=2m}^{2n} |a_j| < \epsilon$ **follow from the Cauchy hypothesis on** $\sum |a_n|$**?:**

Key contrast: convergence (not absolute) is NOT preserved by taking even subsequence; absolute convergence IS.: _____

Pass 4 – Skeleton Proof

(a) State counterexample a_n; verify $\sum a_n$ converges:

(a) Verify $\sum a_{2n}$ diverges; conclude (a) is false:

(b) Fix $\epsilon > 0$; apply Cauchy criterion to $\sum |a_n|$ to get N:

(b) For $n \geq m \geq N$, bound $\sum_{k=m}^{n} |a_{2k}|$:

(b) Conclude $\sum |a_{2n}|$ converges by Cauchy criterion:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 7 – Solutions**

Problem 6.

Solution:

(a) **Disprove:** Consider the series $a_n = \frac{(-1)^n}{n}$. Then:

- $\sum a_n$ converges: by the alternating series test, since $\frac{1}{n}$ is a decreasing sequence that converges to 0.
- $\sum a_{2n} = \sum \frac{1}{2n}$ diverges: this is $\frac{1}{2}$ times the series $\sum \frac{1}{n}$, which we know diverges by lecture.

(b) **Prove:** We will use the Cauchy criterion. Fix $\epsilon > 0$. As $\sum |a_n|$ converges, there exists an N so that

$$\sum_{k=m}^{n} |a_k| < \epsilon \quad \text{for all } n \geq m \geq N.$$

Then for $n \geq m \geq N$, we have

$$\sum_{k=m}^{n} |a_{2k}| \leq \sum_{j=2m}^{2n} |a_j| < \epsilon.$$

(Note that the first sum consists of only even terms $|a_{2m}| + |a_{2m+2}| + \cdots + |a_{2n}|$, which differs from the second sum $|a_{2m}| + |a_{2m+1}| + |a_{2m+2}| + \cdots + |a_{2n}|$ by only the odd terms, each of which is nonnegative.) Therefore $\sum |a_{2n}|$ converges by the Cauchy criterion.$\square$

Reflection & Variation

State the Cauchy criterion for series. Why is it useful here rather than a direct estimate?: _____

In (b), where exactly did the nonnegativity of $|a_j|$ enter?: _____

Can you prove the analogous statement: if $\sum a_n$ converges absolutely, then $\sum a_{2n-1}$ (odd-indexed terms) also converges absolutely?:

_____

What nearby exam variation could appear?: _____

Homework 8

Problem 1

Problem Statement **Homework 8, Problem 1**

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its interior and prove your answer.

(a) $A = [0, 1)$
(b) $A = \mathbb{Z}$
(c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Pass 1 – First Diagnosis

Topic: _____

Best guess for the interior in (a), (b), and (c):: _____

What definition of "interior point" will you use?: _____

Which density facts or neighborhood facts are relevant?: _____

What point should you test first to show a set is not open?: _____

Pass 2 – Proof Recipe Card

(a) Candidate interior: **Why is it an open subset of $[0, 1)$?:** _____

(a) Why is 0 not an interior point? Pick a witness in $(-r, r) \setminus [0, 1)$.: _____

(b) Fix $n \in \mathbb{Z}$. What point in $(n - r, n + r)$ is not an integer?: _____

(c) Fix $a \in A$. What density fact gives a point in $(a - r, a + r) \setminus A$?: _____

How does showing every point fails the interior-point test imply the interior is empty?: _____

Pass 3 – Self-Explanation

Why does $(0, 1) \subseteq A^\circ \subseteq [0, 1)$ follow from the definition of interior?: _____

Why does every open interval around an integer contain nonintegers?: _____

Why does every open interval around a rational in $(0, 1)$ contain irrationals?: _____

What is the exact logical difference between "open subset of A" and "interior point of A"?: _____

Common mistake: showing a set is not open is not enough; you must identify its interior.: _____

Pass 4 – Skeleton Proof

(a) Show $(0,1)$ is an open subset of $[0,1)$; deduce $(0,1) \subseteq [0,1)^\circ$:

(a) Show 0 is not an interior point; conclude $[0,1)^\circ = (0,1)$:

(b) Fix $n \in \mathbb{Z}$ and show every ball around n contains a noninteger:

(b) Conclude $\mathbb{Z}^\circ = \varnothing$:

(c) Fix $a \in A$ and use density of irrationals to find a point outside A:

(c) Conclude $A^\circ = \varnothing$:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 8 – Selected Solutions**

Problem 1.

Solution:

(a) Claim: $A^\circ = (0,1)$. As A° is the largest open subset of A, and $(0,1)$ is one open subset, we must have $(0,1) \subseteq A^\circ \subseteq [0,1)$. Note that 0 is not an interior point: given $r > 0$ we have $(-r, r) \not\subseteq [0,1)$, since $-r/2$ is a point in $(-r, r)$ that is not in $[0,1)$.

(b) Claim: $A^\circ = \varnothing$. Fix $n \in \mathbb{Z}$; we will show that n is not an interior point. Given $r > 0$ we have $(n-r, n+r) \not\subseteq \mathbb{Z}$, since $\min\{n + r/2, n + 1/2\}$ is in $(n-r, n+r)$ but not in $\mathbb{Z}$.

(c) Claim: $A^\circ = \varnothing$. Fix $a \in A$; we will show that a is not an interior point. As the irrational numbers are dense in $\mathbb{R}$, then for any $r > 0$ we can find a point $y \in \mathbb{Q}^c$ between $a - r$ and a. This y is a point in $(a-r, a+r)$ that is not in A, and so $(a-r, a+r) \not\subseteq A$. $\square$

Reflection & Variation

State the definition of $x \in A^\circ$ in a metric space.: _____

Why is it enough in (b) and (c) to show that every point of A fails to be an interior point?: _____

Compare (b) and (c): one uses isolated integers, the other uses density of irrationals. What structural property of each set is killing the interior?:

_____

What nearby exam variation could appear?: _____

Problem 2

Problem Statement **Homework 8, Problem 2**

Recall that d_1 and d_∞ are both metrics on $\mathbb{R}^2$.

(a) Prove that $d_\infty(x,y) \le d_1(x,y) \le 2d_\infty(x,y)$ for all $x, y \in \mathbb{R}^2$.

(b) Prove that for any $x \in \mathbb{R}^2$ and $r > 0$ we have

$$\{y \in \mathbb{R}^2 : d_\infty(x,y) < r/2\} \subseteq \{y \in \mathbb{R}^2 : d_1(x,y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x,y) < r\}.$$

I recommend that you draw a picture of these three sets.

(c) Prove that a set $A \subseteq \mathbb{R}^2$ is open in the metric space $(\mathbb{R}^2, d_1)$ if and only if it is open in the metric space $(\mathbb{R}^2, d_\infty)$.

Pass 1 – First Diagnosis

Topic: _____

What are the formulas for d_1 and d_∞?: _____

Which part is algebraic, which part is geometric, and which part is topological?: _____

What should the picture in part (b) look like?: _____

How will part (b) be used inside part (c)?: _____

Pass 2 – Proof Recipe Card

(a) Show $\max\{|x_1 - y_1|, |x_2 - y_2|\} \le |x_1 - y_1| + |x_2 - y_2|$ because:: _____

(a) Show $|x_1 - y_1| + |x_2 - y_2| \le 2\max\{|x_1 - y_1|, |x_2 - y_2|\}$ because:: _____

(b) If $d_\infty(x,y) < r/2$, then $d_1(x,y) \le 2d_\infty(x,y) <$: _____

(b) If $d_1(x,y) < r$, then $d_\infty(x,y) \le d_1(x,y) <$: _____

(c) To pass openness from one metric to the other, which smaller ball should you place inside the given ball?: _____

Pass 3 – Self-Explanation

Why is $\max\{a,b\} \le a + b$ valid for nonnegative a, b?: _____

Why is $a + b \le 2\max\{a,b\}$ valid for nonnegative a, b?: _____

Why does part (b) give exactly the containments needed to compare open balls?: _____

In part (c), why is it enough to fix $a \in A$ and produce one ball around a

contained in A?: _____

Key lesson: equivalent metrics need not be equal, but they generate the same open sets.: _____

Pass 4 – Skeleton Proof

(a) Fix $x = (x_1, x_2)$ and $y = (y_1, y_2)$; prove $d_\infty(x, y) \leq d_1(x, y)$:

(a) Prove $d_1(x, y) \leq 2d_\infty(x, y)$:

(b) First containment: start from $d_\infty(x, y) < r/2$ and apply part (a):

(b) Second containment: start from $d_1(x, y) < r$ and apply part (a):

(c) Assume A is open in $(\mathbb{R}^2, d_\infty)$; produce a d_1-ball inside A:

(c) Assume A is open in $(\mathbb{R}^2, d_1)$; produce a d_∞-ball inside A:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Problem 2.

Solution:

(a) Fix $x = (x_1, x_2)$ and $y = (y_1, y_2)$ in $\mathbb{R}^2$. Then

$$|x_1 - y_1| \leq |x_1 - y_1| + |x_2 - y_2| = d_1(x, y),$$

$$|x_2 - y_2| \leq |x_1 - y_1| + |x_2 - y_2| = d_1(x, y),$$

and so

$$\max\{|x_1 - y_1|, |x_2 - y_2|\} \leq d_1(x, y).$$

This proves the first inequality: $d_\infty(x, y) \leq d_1(x, y)$. On the other hand, note that

$$|x_1 - y_1| \leq \max\{|x_1 - y_1|, |x_2 - y_2|\} = d_\infty(x, y),$$

$$|x_2 - y_2| \leq \max\{|x_1 - y_1|, |x_2 - y_2|\} = d_\infty(x, y),$$

and so

$$|x_1 - y_1| + |x_2 - y_2| \leq 2d_\infty(x, y).$$

This proves that $d_1(x, y) \leq 2d_\infty(x, y)$.

(b) Let's start with the first containment:

$$\{y \in \mathbb{R}^2 : d_\infty(x, y) < r/2\} \subseteq \{y \in \mathbb{R}^2 : d_1(x, y) < r\}.$$

Fix $y \in \mathbb{R}^2$ so that $d_\infty(x, y) < r/2$. Then by the second inequality in part (a) we have

$$d_1(x, y) \leq 2d_\infty(x, y) < 2 \cdot \frac{r}{2} = r,$$

and so $d_1(x, y) < r$. Next, we turn to the second containment:

$$\{y \in \mathbb{R}^2 : d_1(x, y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x, y) < r\}.$$

Fix $y \in \mathbb{R}^2$ so that $d_1(x, y) < r$. Then by the first inequality in part (a), we have

$$d_\infty(x, y) \leq d_1(x, y) < r,$$

and so $d_\infty(x, y) < r$.

(c) $\Longleftarrow$ Suppose that A is open in $(\mathbb{R}^2, d_\infty)$. Fix $a \in A$. Then there exists $r > 0$ so that

$$B_r^\infty(a) = \{x \in \mathbb{R}^2 : d_\infty(x, a) < r\} \subseteq A.$$

Note that $B_r^1(a) \subseteq B_r^\infty(a)$ by part (b), and so $B_r^1(a) \subseteq B_r^\infty(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in $(\mathbb{R}^2, d_1)$.

$\Longrightarrow$ Suppose that A is open in $(\mathbb{R}^2, d_1)$. Fix $a \in A$. Then there exists $r > 0$ so that $B_r^1(a) \subseteq A$. Note that $B_{r/2}^\infty(a) \subseteq B_r^1(a)$ by part (b), and so $B_{r/2}^\infty(a) \subseteq B_r^1(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in $(\mathbb{R}^2, d_\infty)$. $\square$

Reflection & Variation

Write the definitions of d_1 and d_∞ on $\mathbb{R}^2$.: _____

Why does part (b) immediately imply the two metrics generate the same open sets?: _____

In part (c), where exactly was the factor $1/2$ needed, and why could you not simply reuse the same radius in both directions?:

_____

What nearby exam variation could appear?: _____

Problem 3

Problem Statement **Homework 8, Problem 3**

Let X be a nonempty set, and recall that the discrete metric d is a metric on X. Prove that every set $A \subseteq X$ is open in the metric space (X, d).

Pass 1 – First Diagnosis

Topic: _____

What does the discrete metric equal when $x = y$? When $x \neq y$?: _____

What radius should isolate a single point?: _____

What does an open ball of radius $1/2$ look like?: _____

Why does proving each $a \in A$ is an interior point finish the problem?: _____

Pass 2 – Proof Recipe Card

Fix $a \in A$. What is $B_{1/2}(a)$ in the discrete metric?: _____

Why are the only possible values of $d(x, a)$ equal to 0 and 1?: _____

Why does $B_{1/2}(a) = \{a\} \subseteq A$?: _____

What definition of "open" are you invoking at the last line?: _____

How does the proof work for an arbitrary subset A?: _____

Pass 3 – Self-Explanation

Why is radius $1/2$ a convenient choice rather than radius 1?: _____

Could you replace $1/2$ by any number in $(0, 1)$? Why?: _____

What feature of the discrete metric makes every singleton open?: _____

Why do unions of singletons then suggest that every subset should be open?: _____

Common mistake: forgetting to prove the ball is contained in A, not just nonempty.: _____

Pass 4 – Skeleton Proof

Fix $A \subseteq X$ and $a \in A$; compute $B_{1/2}(a)$ explicitly:

Explain why $d(x, a) < 1/2$ forces $x = a$:

Conclude a is an interior point and hence A is open:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 8 – Selected Solutions**

Problem 3.

Solution:

Fix $A \subseteq X$ and $a \in A$. Notice that

$$B_{1/2}(a) = \{x \in X : d(x, a) < 1/2\} = \{a\}.$$

(Indeed, $d(x, a)$ is either equal to 0 or 1, and so the only time that $d(x, a) < 1/2$ is when $d(x, a) = 0$.) Then $a \in B_{1/2}(a) \subseteq A$, and so a is an interior point for a. As $a \in A$ was arbitrary, we conclude that A is open. $\square$

Reflection & Variation

Why does $B_{1/2}(a) = \{a\}$ in the discrete metric?: _____

Which radii r satisfy $B_r(a) = \{a\}$ for every point a?: _____

Explain why the discrete metric is the "most open" metric possible on a set. What does that mean in terms of open subsets?:

_____

What nearby exam variation could appear?: _____

Problem 4

Problem Statement **Homework 8, Problem 4**

Let (X, d) be any metric space. Define the function $\tilde{d} : X \times X \to \mathbb{R}$ by

$$\tilde{d}(x, y) = \min\{d(x, y), 1\}.$$

(a) Prove that $\tilde{d}$ is also a metric on X.
(b) Prove that a set $A \subseteq X$ is open in (X, d) if and only if it is open in $(X, \tilde{d})$.

Pass 1 – First Diagnosis

Topic: _____

Which four metric axioms must be checked in part (a)?: _____

Where is the triangle inequality for d going to be used?: _____

What is the main idea of part (b): compare d-balls and $\tilde{d}$-balls or compare closures?: _____

Why might you want to choose a radius at most $1/2$ in one direction?: ___

Pass 2 – Proof Recipe Card

(a) Nonnegativity / identity / symmetry are easy because $\tilde{d}(x, y)$ is either : _____

(a) Triangle inequality: what four cases arise depending on whether $d(x, y)$ and $d(y, z)$ are < 1 or ≥ 1?: _____

(b) If $x \in B_r^d(a)$, why does $\tilde{d}(x, a) \leq d(x, a) < r$?: _____

(b) If $x \in B_{r_2}^{\tilde{d}}(a)$ with $r_2 \leq 1/2$, why must $\tilde{d}(x, a) = d(x, a)$?: _____

What radii r_1, r_2 will you choose in the $(X, d) \Rightarrow (X, \tilde{d})$ direction?: _____

Pass 3 – Self-Explanation

Why does truncating distances at 1 preserve nonnegativity, identity, and symmetry?: _____

In the triangle inequality proof, why are the cases with one side ≥ 1 automatically easy?: _____

Why does choosing $r_2 \leq 1/2$ force points in $B_{r_2}^{\tilde{d}}(a)$ to have genuine d-distance $< r_2$?: _____

What does this problem show about changing a metric far away from each

point?: _____

Common mistake: forgetting that $\tilde{d}(x, a) < r_2 \leq 1/2$ rules out the possibility $\tilde{d}(x, a) = 1$.: _____

Pass 4 – Skeleton Proof

(a) Check nonnegativity, identity of indiscernibles, and symmetry:

(a) Triangle inequality: split into four cases and show $\tilde{d}(x,z) \leq \tilde{d}(x,y)+\tilde{d}(y,z)$:

(b) Assume A is open in $(X,\tilde{d})$; show $B_r^d(a) \subseteq B_r^{\tilde{d}}(a) \subseteq A$:

(b) Assume A is open in (X,d); choose $r_2 = \min\{r_1, 1/2\}$ and show $B_{r_2}^{\tilde{d}}(a) \subseteq B_{r_1}^d(a)$:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Problem 4.

Solution:

(a) There are 4 properties to check.

1. $\tilde{d}(x,y) \geq 0$ for all $x, y \in X$. Note that $\tilde{d}(x,y)$ is equal to 1 or $d(x,y)$. Both of these are nonnegative; we know $d(x,y) \geq 0$ since d is a metric on X.
2. $\tilde{d}(x,y) = 0 \iff x = y$. If $x = y$, then $d(x,y) = 0$, so $\tilde{d}(x,y) = \min\{0,1\} = 0$. Conversely, if $\tilde{d}(x,y) = 0$, then $\tilde{d}(x,y) \neq 1$ and so we must have $d(x,y) = \tilde{d}(x,y) = 0$. As d is a metric, this can only mean that $x = y$.
3. $\tilde{d}(x,y) = \tilde{d}(y,x)$ for all $x, y \in X$. As d is a metric, we know that $d(x,y) = d(y,x)$, and so
$$\tilde{d}(x,y) = \min\{d(x,y), 1\} = \min\{d(y,x), 1\} = \tilde{d}(y,x).$$
4. $\tilde{d}(x,z) \leq \tilde{d}(x,y) + \tilde{d}(y,z)$ for all $x, y, z \in X$. We distinguish 4 possible cases:
   - $d(x,y) < 1$ and $d(y,z) < 1$: Then
   $$\tilde{d}(x,z) \leq d(x,z) \leq d(x,y) + d(y,z) = \tilde{d}(x,y) + \tilde{d}(y,z).$$
   - $d(x,y) < 1$ and $d(y,z) \geq 1$: Then
   $$\tilde{d}(x,z) \leq 1 \leq d(x,y) + 1 = \tilde{d}(x,y) + \tilde{d}(y,z).$$
   - $d(x,y) \geq 1$ and $d(y,z) < 1$: Then
   $$\tilde{d}(x,z) \leq 1 \leq 1 + d(y,z) = \tilde{d}(x,y) + \tilde{d}(y,z).$$
   - $d(x,y) \geq 1$ and $d(y,z) \geq 1$: Then
   $$\tilde{d}(x,z) \leq 1 \leq 1 + 1 = \tilde{d}(x,y) + \tilde{d}(y,z).$$

   In all cases, the triangle inequality holds.

(b) $\Longleftarrow$ Suppose that A is open in $(X, \tilde{d})$. Fix $a \in A$. Then there exists $r > 0$ so that
$$B_r^{\tilde{d}}(a) = \{x \in X : \tilde{d}(x,a) < r\} \subseteq A.$$
Note that $B_r^d(a) \subseteq B_r^{\tilde{d}}(a)$, since
$$x \in B_r^d(a) \Longrightarrow \tilde{d}(x,a) \leq d(x,a) < r \Longrightarrow x \in B_r^{\tilde{d}}(a).$$
So $B_r^d(a) \subseteq B_r^{\tilde{d}}(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in (X, d).

$\Longrightarrow$ Suppose that A is open in (X, d). Fix $a \in A$. Then there exists $r_1 > 0$ so that $B_{r_1}^d(a) \subseteq A$. Set $r_2 = \min\{r_1, 1/2\}$. Note that $B_{r_2}^{\tilde{d}}(a) \subseteq B_{r_1}^d(a)$, since
$$x \in B_{r_2}^{\tilde{d}}(a) \Longrightarrow \tilde{d}(x,a) = \min\{d(x,a), 1\} < r_2 \leq 1/2$$
$$\Longrightarrow \tilde{d}(x,a) = d(x,a) \Longrightarrow d(x,a) < r_2 \leq r_1.$$
So $B_{r_2}^{\tilde{d}}(a) \subseteq B_{r_1}^d(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in $(X, \tilde{d})$. $\square$

Reflection & Variation

Which metric axiom is the only genuinely nontrivial part of (a)?: _____

Why does truncating distances at 1 leave all open sets unchanged?: _____

Could you replace 1 by any constant $M > 0$ in the definition $\tilde{d}(x, y) = \min\{d(x, y), M\}$? Which parts of the proof would change?:

_____

What nearby exam variation could appear?: _____

Problem 5

Problem Statement **Homework 8, Problem 5**

Let (X, d) be a metric space, $A \subseteq X$, and $x \in X$. Prove that x is in A° if and only if there exists an open set U so that $x \in U \subseteq A$.

Pass 1 – First Diagnosis

Topic: _____

What does $x \in A^\circ$ mean by definition?: _____

What open set should you choose in the forward direction?: _____

What fact about open sets will you use in the reverse direction?: _____

Why is this really just a reformulation of the definition of interior?: _____

Pass 2 – Proof Recipe Card

$\Longrightarrow$ **If $x \in A^\circ$, then there exists $r > 0$ such that .....:** _____

$\Longrightarrow$ **Choose $U =$ Why is U open?:** _____

$\Longleftarrow$ **If $x \in U \subseteq A$ and U is open, then there exists $r > 0$ such that ...:** _____

$\Longleftarrow$ **Combine $B_r(x) \subseteq U$ with $U \subseteq A$ to get .....:** _____

Final conclusion about interior points:: _____

Pass 3 – Self-Explanation

Why is an open ball always an open set in a metric space?: _____

Why does $x \in U$ and "U open" imply the existence of some ball around x contained in U?: _____

What is the difference between proving $x \in A$ and proving $x \in A^\circ$?: _____

How does this problem repackage the definition of interior in a more flexible form?: _____

Common mistake: taking $U = A$ in the forward direction without knowing A is open.: _____

Pass 4 – Skeleton Proof

$\Longrightarrow$ **Assume $x \in A^\circ$; choose $r > 0$ with $B_r(x) \subseteq A$, and set $U = \ldots$**

$\Longleftarrow$ **Assume $x \in U \subseteq A$ with U open; produce $r > 0$ with $B_r(x) \subseteq U$:**

Combine containments to conclude $x \in A^\circ$:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 8 – Selected Solutions**

Problem 5.

Solution:

$\Longrightarrow$ Suppose that $x \in A^\circ$. Then there exists $r > 0$ such that $B_r(x) \subseteq A$. Notice that $U = B_r(x)$ is an open set that satisfies $x \in U \subseteq A$.

$\Longleftarrow$ Suppose that $x \in U \subseteq A$ for some open set U. As U is open, there exists $r > 0$ so that $B_r(x) \subseteq U$. Together we have $B_r(x) \subseteq U \subseteq A$, and so $x \in A^\circ$. $\square$

Reflection & Variation

State the definition of A° directly in terms of open balls.: _____

Why is Problem 5 an "if and only if" restatement of that definition?: ____

Suppose U is open and $U \subseteq A$. What does Problem 5 say about every point of U relative to A°?:

_____

What nearby exam variation could appear?: _____

Problem 6

Problem Statement **Homework 8, Problem 6**

Let (X, d) be a metric space and let A be a nonempty subset of X. Prove that A is open if and only if there exists a set I, positive numbers $r_i > 0$, and points $a_i \in X$ so that

$$A = \bigcup_{i \in I} B_{r_i}(a_i).$$

Pass 1 – First Diagnosis

Topic: _____

What theorem about unions of open sets do you expect to use?: _____

If A is open, what natural index set I can you choose?: _____

If A is a union of open balls, why should A be open?: _____

What role does the assumption $A \neq \varnothing$ play?: _____

Pass 2 – Proof Recipe Card

$\Longrightarrow$ **For each $a \in A$, openness gives an $r_a > 0$ such that .....:** _____

$\Longrightarrow$ **Choose $I =$ and define a_i, r_i how?:** _____

$\Longrightarrow$ **Why does $A \subseteq \bigcup_{i \in I} B_{r_i}(a_i)$ hold?:** _____

$\Longleftarrow$ **Why is each ball $B_{r_i}(a_i)$ open?:** _____

$\Longleftarrow$ **Which closure property of open sets finishes the proof?:** _____

Pass 3 – Self-Explanation

Why is every point of an open set contained in some open ball lying inside the set?: _____

How does choosing one ball around each point convert a local definition into a global union formula?: _____

Why is an arbitrary union of open sets open?: _____

Could you take $I = A$ in the forward direction? Explain the notation carefully.: _____

Common mistake: proving only one containment in the union identity.: _____

Pass 4 – Skeleton Proof

$\Longrightarrow$ **Assume A is open; for each $a \in A$, choose $r_a > 0$ with $B_{r_a}(a) \subseteq A$:**

Show $A = \bigcup_{a \in A} B_{r_a}(a)$ by proving both containments:

$\Longleftarrow$ **Assume $A = \bigcup_{i \in I} B_{r_i}(a_i)$; explain why each ball is open:**

Use closure of openness under unions to conclude A is open:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 8 – Selected Solutions**

Problem 6.

Solution:

No official solution provided.

Reflection & Variation

What is the difference between a local openness condition and a global union-of-balls description?: _____

Why is the forward direction essentially "one ball per point"?: _____

If A is open, can you always write it as a union of balls centered at points of A rather than arbitrary points of X? Explain.:

_____

What nearby exam variation could appear?: _____

Homework 9

Problem 1

Problem Statement **Homework 9, Problem 1**

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its closure, isolated points, and limit points. Prove your answer.

(a) $A = [0, 1)$
(b) $A = \mathbb{Z}$
(c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Pass 1 – First Diagnosis

Topic: _____

Best guess for the closure, isolated points, and limit points in (a), (b), and (c):: _____

Which definitions do you need: adherent point, isolated point, limit point?: _____

Which density facts are relevant?: _____

What endpoints or special points should you test first?: _____

Pass 2 – Proof Recipe Card

(a) Why is 1 in $\overline{A}$? Find a point of $(1 - r, 1 + r) \cap A$.: _____

(a) Why is no point of $[0, 1)$ isolated? Find another point of A in $(a-r, a+r)$.:

(b) Why is $x \notin \overline{\mathbb{Z}}$ for $x \in \mathbb{Z}^c$? Choose a radius inside one gap.: _____

(b) Why is every integer isolated?: _____

(c) How do density of rationals and density of irrationals determine closure and isolated points?: _____

Pass 3 – Self-Explanation

Why does closure ask for every ball to meet A, while isolated asks for one ball to meet A only at the point?: _____

Why is $\mathbb{Z}$ closed but made entirely of isolated points?: _____

Why is $\{x \in \mathbb{Q} : 0 < x < 1\}$ dense in $[0, 1]$ but has no isolated points?: _____

Where is the decomposition into isolated points and limit points being used?: _____

Common mistake: confusing "point of A" with "point of $\overline{A}$".: _____

Pass 4 – Skeleton Proof

(a) Show $\overline{A} = [0, 1]$ by proving 1 is adherent and nothing outside $[0, 1]$ is needed:

(a) Show isolated points are empty and conclude limit points:

(b) Show $\overline{\mathbb{Z}} = \mathbb{Z}$ and each integer is isolated:

(b) Conclude limit points are empty:

(c) Show $\overline{A} = [0, 1]$ using density, then show isolated points are empty:

(c) Conclude limit points:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 9 – Solutions**

Problem 1.

Solution:

(a) Closure: $\overline{A} = [0,1]$. As $\overline{A}$ is the smallest closed set containing A, and $[0,1]$ is one such closed set, we must have $[0,1) \subseteq \overline{A} \subseteq [0,1]$. Note that 1 must be in the closure of A: given $r > 0$ we have $(1-r, 1+r) \cap [0,1) \neq \varnothing$, since $\max\{1-r/2, 0\}$ is in $(1-r, 1+r)$ and $[0,1)$.

Isolated points: $\varnothing$. Fix $a \in A$. Given $r > 0$ we have $(a-r, a+r) \cap A \neq \{a\}$, since $\min\{a + r/2, (a+1)/2\}$ is a point in $(a-r, a+r) \cap A$ other than a.

Limit points: $A' = [0,1]$. By definition, $\overline{A}$ is the union of isolated points and limit points, and we just showed that there are no isolated points.

(b) Closure: $\overline{A} = \mathbb{Z}$. We automatically have $\overline{A} \supseteq \mathbb{Z}$. Fix $x \in \mathbb{Z}^c$; we will show that x is not an adherent point. Let $n \in \mathbb{Z}$ so that $n < x < n+1$. Then $r = \min\{x-n, n+1-x\}$ is a positive number so that $(x-r, x+r) \cap \mathbb{Z} = \varnothing$.

Isolated points: $\mathbb{Z}$. Fix $n \in \mathbb{Z}$. Then $(n-1, n+1) \cap \mathbb{Z} = \{n\}$.

Limit points: $A' = \varnothing$. By definition, $\overline{A}$ is the union of isolated points and limit points, and we just showed that every point is an isolated point.

(c) Closure: $\overline{A} = [0,1]$. Note that 0 is an adherent point for A: given $r > 0$, pick $n > \max\{1/r, 1\}$ and then $1/n$ is a point in $(-r, r) \cap A$. Now fix $x \in (0,1]$; we will show that x is an adherent point. As the rational numbers are dense in $\mathbb{R}$, given $r > 0$ we can find a point $q \in \mathbb{Q}$ between $\max\{x-r, 0\}$ and x, and this point q will be in $(x-r, x+r) \cap A$.

Isolated points: $\varnothing$. Given $x \in A$ and $r > 0$, we pick $q \in \mathbb{Q}$ as in the previous paragraph. This q is a point in $(x-r, x+r) \cap A$ other than x.

Limit points: $A' = [0,1]$. By definition, $\overline{A}$ is the union of isolated points and limit points, and we just showed that there are no isolated points. $\square$

Reflection & Variation

State the definitions of adherent point, isolated point, and limit point.: ___

Why can the closure be larger than the set even when the set has no isolated points?: _____

Compare parts (b) and (c): one is closed and discrete, the other is dense in an interval. How do those structural differences control the isolated and limit points?:

What nearby exam variation could appear?: _____

Problem 2

Problem Statement **Homework 9, Problem 2**

Let (X, d) be a metric space, let A be a nonempty subset of X, and let U be an open subset of X.

(a) Prove that $U \cap \overline{A} \subseteq \overline{U \cap A}$.

(b) Prove that $\overline{U \cap \overline{A}} = \overline{U \cap A}$. (Hint: Use part (a) and properties of closure from lecture.)

(c) Conclude that if $U \cap A = \varnothing$, then $U \cap \overline{A} = \varnothing$.

Pass 1 – First Diagnosis

Topic: _____

What does it mean to show $x \in \overline{U \cap A}$?: _____

Where will the openness of U be used?: _____

What closure properties from lecture are relevant for part (b)?: _____

How should part (c) follow once part (b) is known?: _____

Pass 2 – Proof Recipe Card

(a) Fix $x \in U \cap \overline{A}$. To prove $x \in \overline{U \cap A}$, fix $r > 0$ and aim to show ....: _____

(a) Since U is open and $x \in U$, choose $s > 0$ with .....: _____

(a) Set $\tilde{r} = \min\{r, s\}$. Since $x \in \overline{A}$, what point y can you choose?: _____

(b) Use $B \subseteq C \Rightarrow \overline{B} \subseteq \overline{C}$ on which inclusion?: _____

(c) Why does $\overline{B} = \varnothing$ force $B = \varnothing$?: _____

Pass 3 – Self-Explanation

Why does the openness of U let you shrink an arbitrary r to a smaller radius inside U?: _____

Why is $y \in B_{\tilde{r}}(x)$ enough to conclude both $y \in B_r(x)$ and $y \in U$?: _____

In part (b), where exactly do you use $A \subseteq \overline{A}$?: _____

Why does part (c) express the idea that open sets disjoint from A are also disjoint from $\overline{A}$?: _____

Common mistake: confusing $U \cap \overline{A}$ with $\overline{U \cap A}$; part (a) proves only one inclusion.: _____

Pass 4 – Skeleton Proof

(a) Fix $x \in U \cap \overline{A}$ and $r > 0$; choose $s > 0$ with $B_s(x) \subseteq U$:

Set $\tilde{r} = \min\{r, s\}$ and pick $y \in B_{\tilde{r}}(x) \cap A$; verify $y \in B_r(x) \cap U \cap A$:

(b) Apply closure monotonicity to part (a) and use $A \subseteq \overline{A}$:

Conclude $\overline{U \cap \overline{A}} = \overline{U \cap A}$:

(c) Start from $U \cap A = \varnothing$ and use part (b):

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Problem 2.

Solution:

(a) Fix $x \in U \cap \overline{A}$. We want to show that $x \in \overline{U \cap A}$, i.e., for any $r > 0$, $B_r(x) \cap U \cap A$ is nonempty.

Fix $r > 0$. As $x \in U$ and U is open, there exists $s > 0$ so that $B_s(x) \subseteq U$. Set $\tilde{r} = \min\{r, s\}$. As $x \in \overline{A}$ and $\tilde{r}$ is a positive number, then we know that $B_{\tilde{r}}(x) \cap A$ is nonempty, and so there exists some point $y \in B_{\tilde{r}}(x) \cap A$.

Altogether, we have:

- $y \in A$
- $y \in B_r(x)$: since $y \in B_{\tilde{r}}(x)$, and so $d(y, x) < \tilde{r} \leq r$ and thus $y \in B_r(x)$.
- $y \in U$: since $y \in B_{\tilde{r}}(x)$, and so $d(y, x) < \tilde{r} \leq s$ and thus $y \in B_s(x) \subseteq U$.

So y is a point in $B_r(x) \cap U \cap A$, as desired.

(b) Recall that $B \subseteq C \implies \overline{B} \subseteq \overline{C}$. Applying this to part (a), we see that

$$\overline{U \cap \overline{A}} \subseteq \overline{\overline{U \cap A}}.$$

On the other hand, we have $A \subseteq \overline{A}$ and so $U \cap A \subseteq U \cap \overline{A}$. Therefore

$$\overline{U \cap A} \subseteq \overline{U \cap \overline{A}}.$$

Together, this shows that $\overline{U \cap \overline{A}} = \overline{U \cap A}$.

(c) For any set B, we have $B \subseteq \overline{B}$. So, if $\overline{B} = \varnothing$, then we must have $B = \varnothing$. Therefore:

$$\begin{aligned} U \cap A = \varnothing &\implies \overline{U \cap A} = \varnothing \\ &\implies \overline{U \cap \overline{A}} = \varnothing \qquad \text{by (b)} \\ &\implies U \cap \overline{A} = \varnothing. \square \end{aligned}$$

Reflection & Variation

Where exactly was the openness of U used in part (a)?: _____

State the monotonicity property of closure used in part (b).: _____

Explain part (c) in words: if an open set misses A, why must it also miss $\overline{A}$?:

_____

What nearby exam variation could appear?: _____

Problem 3

Problem Statement **Homework 9, Problem 3**

Recall that d_2 is a metric on $\mathbb{R}^2$. Consider the subspace

$$Y = \{(x, 0) : x \in \mathbb{R}\}$$

with the induced metric d_Y.

(a) Provide an example of a set $A \subseteq Y$ where A is open in (Y, d_Y) but not open in $(\mathbb{R}^2, d_2)$.

(b) Prove that for any $B \subseteq Y$, if B is closed in (Y, d_Y) then B is closed in $(\mathbb{R}^2, d_2)$.

Pass 1 – First Diagnosis

Topic: _____

What geometric set is Y inside $\mathbb{R}^2$?: _____

What kind of subset of the x-axis is a natural candidate for part (a)?: ____

Why should that candidate fail to be open in $\mathbb{R}^2$?: _____

What lecture fact about closed subspaces is relevant in part (b)?: _____

Pass 2 – Proof Recipe Card

(a) Choose $A = \{(x, 0) : \ldots\ldots\ldots\}$. Why is it open in the subspace?: _____

(a) To show it is not open in $\mathbb{R}^2$, which point of A should you test?: _____

(a) What point in the Euclidean ball around $(0, 0)$ lies off the x-axis?: ____

(b) Which corollary lets you reduce the claim to showing that Y is closed in $\mathbb{R}^2$?: _____

(b) For $x = (x_1, x_2) \in Y^c$, what radius makes $B_r^{\mathbb{R}^2}(x) \subseteq Y^c$?: _____

Pass 3 – Self-Explanation

Why does a horizontal interval on the x-axis look open inside Y?: _____

Why is openness in the subspace weaker than openness in the ambient space?: _____

How does the second coordinate detect whether a point lies in Y^c?: _____

Why does proving Y^c is open imply Y is closed?: _____

Common mistake: confusing "open in Y" with "open in $\mathbb{R}^2$".: _____

Pass 4 – Skeleton Proof

(a) State the example A and prove it is open in (Y, d_Y) by choosing a radius around $(x_1, 0)$:

(a) Show $(0,0) \in A$ is not interior in $\mathbb{R}^2$ by finding a point off the axis:

(b) Show Y is closed by proving Y^c is open: fix $(x_1, x_2) \in Y^c$ and choose $r = |x_2|$:

Apply the subspace corollary to conclude the claim about closed subsets $B \subseteq Y$:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Problem 3.

Solution:

(a) Consider the set $A = \{(x, 0) : -1 < x < 1\}$.

First, we will show that A is open in (Y, d_Y). Fix $x = (x_1, 0) \in A$. Set $r = 1 - |x_1|$. Then $r > 0$ and

$$B_r^Y(x) = \{y \in Y : d_Y(x, y) < r\} \subseteq A,$$

since

$$\begin{aligned} y \in B_r^Y(x) &\Longrightarrow r > d_Y(x, y) = \sqrt{|x_1 - y_1|^2 + |0 - 0|^2} = |x_1 - y_1| \\ &\Longrightarrow |y_1| \leq |x_1| + |y_1 - x_1| < |x_1| + r = 1 \\ &\Longrightarrow y \in A. \end{aligned}$$

Next, we will show that A is not open in $(\mathbb{R}^2, d_2)$ by demonstrating that $(0, 0) \in A$ is not an interior point. Fix $r > 0$. Then

$$B_r^{\mathbb{R}^2}((0, 0)) = \{y \in \mathbb{R}^2 : d_2(y, (0, 0)) < r\} \not\subseteq A,$$

since $(0, r/2)$ is a point that is in $B_r^{\mathbb{R}^2}((0, 0))$ but not in A.

(b) By the corollary from Lecture 24, it suffices to show that Y is closed in $(\mathbb{R}^2, d_2)$. Fix $x = (x_1, x_2) \in Y^c$. Then $x_2 \neq 0$. Set $r = |x_2|$. Then $r > 0$ and

$$B_r^{\mathbb{R}^2}(x) = \{y \in \mathbb{R}^2 : d_2(x, y) < r\} \subseteq Y^c,$$

since

$$\begin{aligned} y \in B_r^{\mathbb{R}^2}(x) &\Longrightarrow r > d_2(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2} \geq |x_2 - y_2| \\ &\Longrightarrow |y_2| \geq |x_2| - |x_2 - y_2| > |x_2| - r = 0 \\ &\Longrightarrow y_2 \neq 0 \Longrightarrow y \in Y^c. \end{aligned}$$

As $x \in Y^c$ was arbitrary, we conclude that Y^c is open, and so Y is closed. $\square$

Reflection & Variation

What is the induced metric d_Y measuring on the x-axis?: _____

Why is the interval $(-1, 1)$ along the axis open in the subspace but not in the plane?: _____

Explain the role of the corollary from Lecture 24 in part (b). Why is proving Y closed enough to control closed sets inside the subspace?:

What nearby exam variation could appear?: _____

Problem 4

Problem Statement **Homework 9, Problem 4**

Let (X, d) be a metric space, $\{x_n\}_{n\geq 1}$ a sequence in X, and $x \in X$. Prove that the sequence $\{x_n\}_{n\geq 1}$ converges to x if and only if for every open set U that contains x, we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$.

Pass 1 – First Diagnosis

Topic: _____

What does convergence to x mean in terms of ϵ-balls?: _____

What does "for all but finitely many n" mean precisely?: _____

How will the openness of U be used in the forward direction?: _____

Which open set should you test in the reverse direction?: _____

Pass 2 – Proof Recipe Card

$\Longrightarrow$ **If $x_n \to x$ and U is open with $x \in U$, choose $r > 0$ so that .....:** _____

$\Longrightarrow$ **Convergence gives N such that .....:** _____

$\Longleftarrow$ **Fix $r > 0$ and apply the hypothesis to the open set .....:** _____

$\Longleftarrow$ **If only finitely many indices miss $B_r(x)$, how do you choose N?:** _____

Why does this prove the usual definition of convergence?: _____

Pass 3 – Self-Explanation

Why does every open neighborhood of x contain some open ball centered at x?: _____

Why is "all but finitely many" the right topological substitute for "eventually"?: _____

In the reverse direction, why is it enough to test the neighborhood $B_r(x)$ for arbitrary $r > 0$?: _____

What is the conceptual statement of this problem in one sentence?: _____

Common mistake: forgetting that the index N may depend on the chosen open set U or radius r.: _____

Pass 4 – Skeleton Proof

$\Longrightarrow$ **Fix an open set $U \ni x$; choose $r > 0$ with $B_r(x) \subseteq U$, then apply convergence:**

$\Longleftarrow$ **Fix $r > 0$ and apply the hypothesis to $B_r(x)$; choose N beyond the bad indices:**

Conclude $d(x_n, x) < r$ for all $n \geq N$, and hence $x_n \to x$:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 9 – Solutions**

Problem 4.

Solution:

$\Longrightarrow$ Suppose that $x_n \to x$ in (X, d) as $n \to \infty$. Fix an open set $U \subseteq X$ such that $x \in U$. As U is open, then there exists $r > 0$ so that $B_r(x) \subseteq U$. Then, by definition of convergence, we know that there exists an N so that $d(x_n, x) < r$ for all $n \geq N$. Together, we conclude that $x_n \in B_r(x) \subseteq U$ for all $n \geq N$.

$\Longleftarrow$ Suppose that for any open set containing x, we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$. Fix $r > 0$. Applying this fact to the open set $B_r(x)$, we see that $x_n \in B_r(x)$ for all but finitely many $n \in \mathbb{N}$. Let N be larger than the largest index for which $x_n \notin B_r(x)$. Then, for any $n \geq N$, we have $x_n \in B_r(x)$ and thus $d(x_n, x) < r$. As $r > 0$ was arbitrary, we conclude that $\{x_n\}_{n \geq 1}$ converges to x in (X, d). $\square$

Reflection & Variation

Translate "for all but finitely many n" into a statement using a single index N.: _____

Why is this result the topological version of metric convergence?: _____

If x_n fails to converge to x, what kind of open neighborhood of x must capture infinitely many failures? Explain using the contrapositive.:

_____

What nearby exam variation could appear?: _____

Problem 5

Problem Statement **Homework 9, Problem 5**

Recall that d_2 is a metric on $\mathbb{R}^2$. Suppose that $x_n = (x_n^1, x_n^2)$ is a sequence of points in $\mathbb{R}^2$.

(a) Prove that $|x_n^1 - x_1| \leq d_2(x_n, x)$ for any $x = (x_1, x_2) \in \mathbb{R}^2$.

(b) Prove that the sequence $\{x_n\}_{n\geq 1}$ converges in $(\mathbb{R}^2, d_2)$ if and only if the sequences $\{x_n^1\}_{n\geq 1}$ and $\{x_n^2\}_{n\geq 1}$ of real numbers both converge.

Pass 1 – First Diagnosis

Topic: _____

What is the formula for $d_2(x_n, x)$?: _____

Why should one coordinate difference be bounded by the full Euclidean distance?: _____

What theorem will part (a) feed into in part (b)?: _____

How do limit laws enter the reverse direction of part (b)?: _____

Pass 2 – Proof Recipe Card

(a) Write $d_2(x_n, x) = \left(|x_n^1 - x_1|^2 + |x_n^2 - x_2|^2\right)^{1/2}$ and compare to .....: _____

(b) If $x_n \to x$, then $d_2(x_n, x) \to 0$. Use part (a) to show .....: _____

(b) Which theorem turns $0 \leq |x_n^1 - x_1| \leq d_2(x_n, x)$ into coordinate convergence?: _____

(b) If $x_n^1 \to x_1$ and $x_n^2 \to x_2$, what happens to $|x_n^1 - x_1|^2 + |x_n^2 - x_2|^2$?: _____

How do you conclude $d_2(x_n, x) \to 0$?: _____

Pass 3 – Self-Explanation

Why is adding a nonnegative term inside a square root enough to make the whole quantity larger?: _____

Why does convergence in $\mathbb{R}^2$ imply coordinatewise convergence?: _____

Why does coordinatewise convergence imply convergence in $\mathbb{R}^2$ specifically for the Euclidean metric?: _____

Where do you use the squeeze theorem, and where do you use limit laws?:

Common mistake: proving only one coordinate converges in the forward direction.: _____

Pass 4 – Skeleton Proof

(a) Expand $d_2(x_n, x)$ and compare it to $|x_n^1 - x_1|$:

(b) Assume $x_n \to x$; use part (a) and squeeze theorem for both coordinates:

(b) Assume coordinatewise convergence; apply limit laws to $d_2(x_n, x)$:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Problem 5.

Solution:

(a) Recall that

$$d_2(x_n, x) = \left(|x_n^1 - x_1|^2 + |x_n^2 - x_2|^2\right)^{1/2}.$$

As $|x_n^2 - x_2|^2 \geq 0$, then

$$d_2(x_n, x) \geq \left(|x_n^1 - x_1|^2 + 0\right)^{1/2} \geq |x_n^1 - x_1|.$$

(b) $\Longrightarrow$ Suppose that $\{x_n\}_{n\geq 1}$ converges to x in $(\mathbb{R}^2, d_2)$. Then

$$d_2(x_n, x) \to 0 \quad \text{as } n \to \infty.$$

By part (a), we know that

$$0 \leq |x_n^1 - x_1| \leq d_2(x_n, x).$$

So, by the squeeze theorem we have

$$|x_n^1 - x_1| \to 0 \quad \text{as } n \to \infty,$$

and so $\{x_n^1\}_{n\geq 1}$ converges to x_1 in $(\mathbb{R}, |\cdot|)$. Similarly, we have

$$0 \leq |x_n^2 - x_2| \leq d_2(x_n, x),$$

and so $\{x_n^2\}_{n\geq 1}$ converges to x_2 in $(\mathbb{R}, |\cdot|)$.

$\Longleftarrow$ Suppose that $\{x_n^1\}_{n\geq 1}$ converges to x_1 and $\{x_n^2\}_{n\geq 1}$ converges to x_2. Then

$$|x_n^1 - x_1| \to 0 \quad \text{and} \quad |x_n^2 - x_2| \to 0 \quad \text{as } n \to \infty.$$

So, by the limit laws we have

$$d_2(x_n, x) = \left(|x_n^1 - x_1|^2 + |x_n^2 - x_2|^2\right)^{1/2} \to 0 \quad \text{as } n \to \infty,$$

and thus $\{x_n\}_{n\geq 1}$ converges to x in $(\mathbb{R}^2, d_2)$. $\square$

Reflection & Variation

Why does the Euclidean distance dominate each coordinate difference?: __

Where did the squeeze theorem appear in the proof?: _____

Would the same "coordinatewise iff metric" convergence statement hold

for d_1 and d_∞ on $\mathbb{R}^2$? Explain briefly.:

_____

What nearby exam variation could appear?: _____

Problem 6

Problem Statement **Homework 9, Problem 6**

Let (X, d) be a metric space and $\{x_n\}_{n\geq 1}$ a sequence in X. Prove that if $\{x_n\}_{n\geq 1}$ converges to $x \in X$, then for any $y \in X$, the sequence $\{d(x_n, y)\}_{n\geq 1}$ of real numbers converges to $d(x, y)$.

Pass 1 – First Diagnosis

Topic: _____

What inequality involving $d(x_n, y)$ and $d(x, y)$ should come from triangle inequality?: _____

What is the "reverse triangle inequality" in this setting?: _____

How does convergence of x_n to x enter the final squeeze step?: _____

What real sequence are you proving converges?: _____

Pass 2 – Proof Recipe Card

Apply triangle inequality to $d(x_n, y)$ using the point x:: _____

Apply triangle inequality to $d(x, y)$ using the point x_n:: _____

Combine the two inequalities to obtain .....: _____

Why does the right-hand side converge to 0?: _____

Which theorem finishes the proof once you have an absolute-value bound?:

Pass 3 – Self-Explanation

Why do you need two applications of triangle inequality rather than one?:

How do those two inequalities turn into an absolute-value estimate?: _____

What does the result say conceptually about the map $z \mapsto d(z, y)$?: _____

Why is this a continuity statement?: _____

Common mistake: writing $d(x_n, y) \to d(x, y)$ without first bounding the difference.: _____

Pass 4 – Skeleton Proof

Apply triangle inequality to bound $d(x_n, y) - d(x, y)$ above:

Apply triangle inequality again to bound $d(x, y) - d(x_n, y)$ above:

Combine into $|d(x_n, y) - d(x, y)| \leq d(x_n, x)$ and conclude by squeeze theorem:

Pass 5 – Full Proof Attempt

Write a complete proof before reading the Official Solution.

Official Solution **Homework 9 – Solutions**

Problem 6.

Solution:

Fix $y \in X$. By triangle inequality,

$$d(x_n, y) \leq d(x_n, x) + d(x, y) \Longrightarrow d(x_n, y) - d(x, y) \leq d(x_n, x)$$

$$d(x, y) \leq d(x, x_n) + d(x_n, y) \Longrightarrow d(x, y) - d(x_n, y) \leq d(x_n, x)$$

Together, we obtain

$$|d(x_n, y) - d(x, y)| \leq d(x_n, x).$$

As $\{x_n\}_{n\geq 1}$ converges to x in (X, d), then the right-hand side above converges to 0 as $n \to \infty$. Therefore, by the squeeze theorem, we conclude that

$$|d(x_n, y) - d(x, y)| \to 0 \quad \text{as } n \to \infty,$$

and thus $d(x_n, y)$ converges to $d(x, y)$. $\square$

Reflection & Variation

State the reverse triangle inequality in a metric space.: _____

Why does this problem show that $z \mapsto d(z, y)$ is continuous?: _____

How would you adapt the proof to show $|d(x_n, y_n) - d(x, y)| \leq d(x_n, x) + d(y_n, y)$ if also $y_n \to y$?:

_____

What nearby exam variation could appear?: _____

Practice Midterm 2

Problems and solutions to be added.